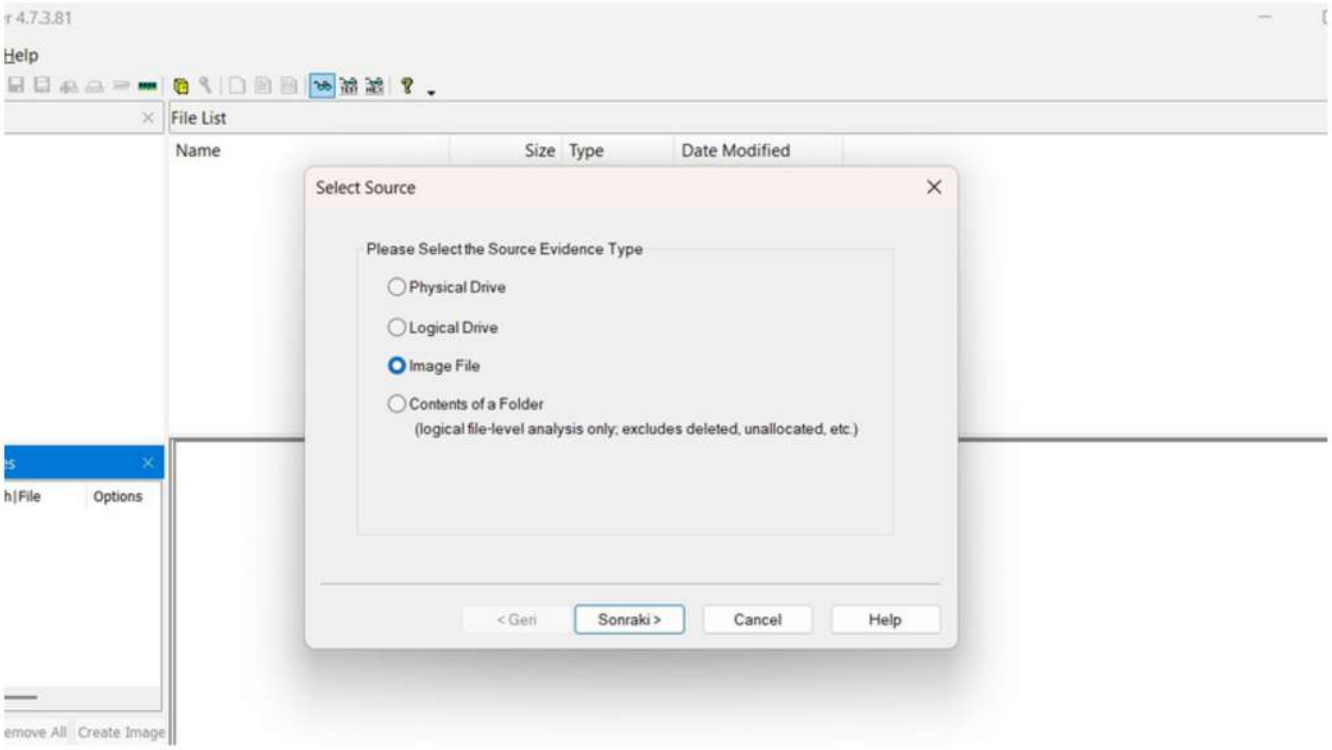
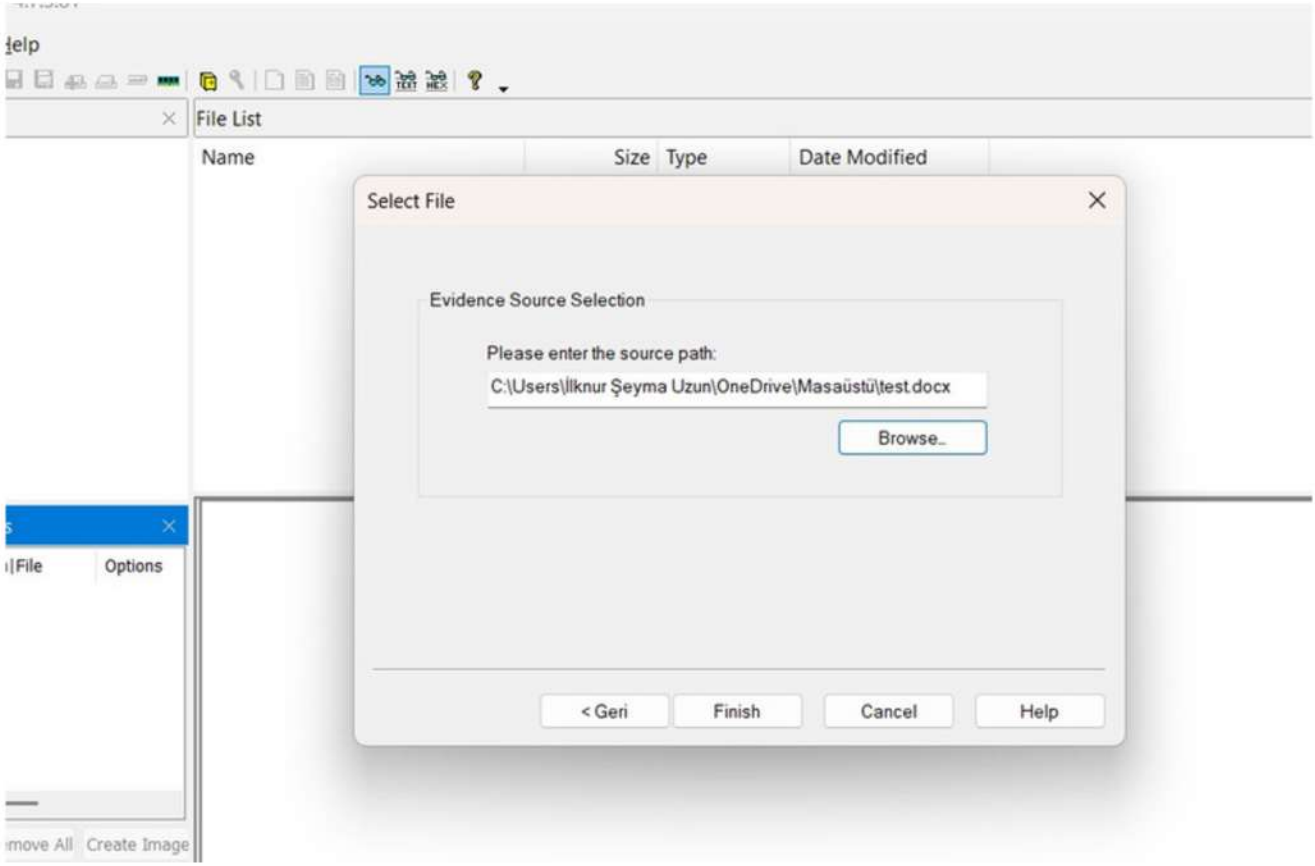
Hash Değeri Alma

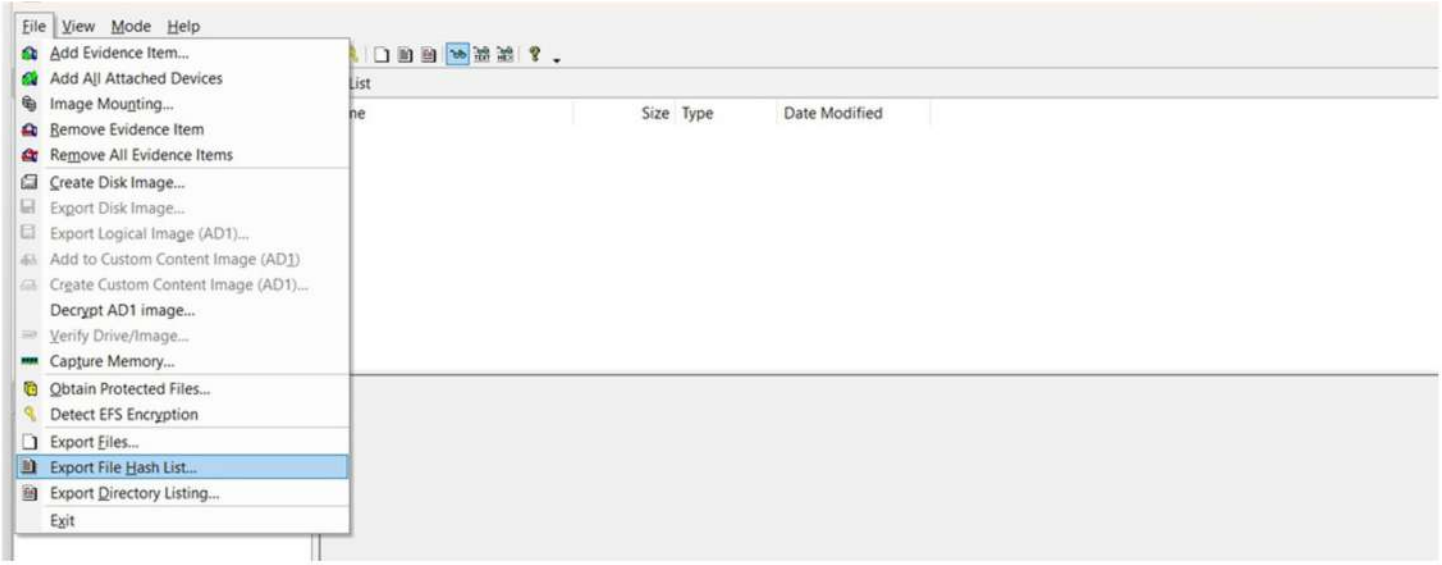
Hash dosyanın içeriğinin dijital parmak izidir. Dosya değişirse hash değeri de değişir. Bu sayede veri bütünlüğü kontrol edilir. ilk önce dosyanın hash değeri alınır böylece delilde herhangi bir değişiklik yapılıp yapılmadığı ispatlanabilir.

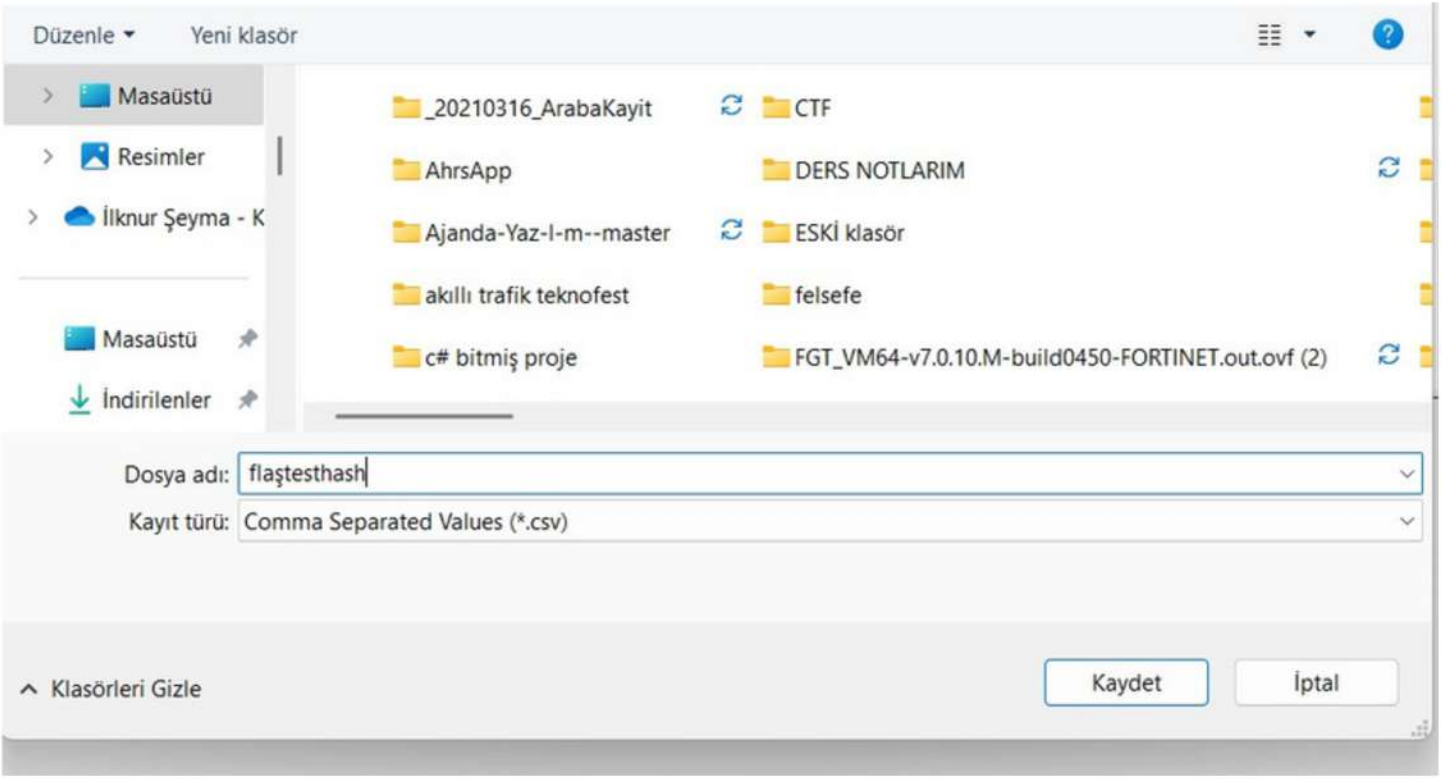
Imaj dosyasından Hash alma adımları

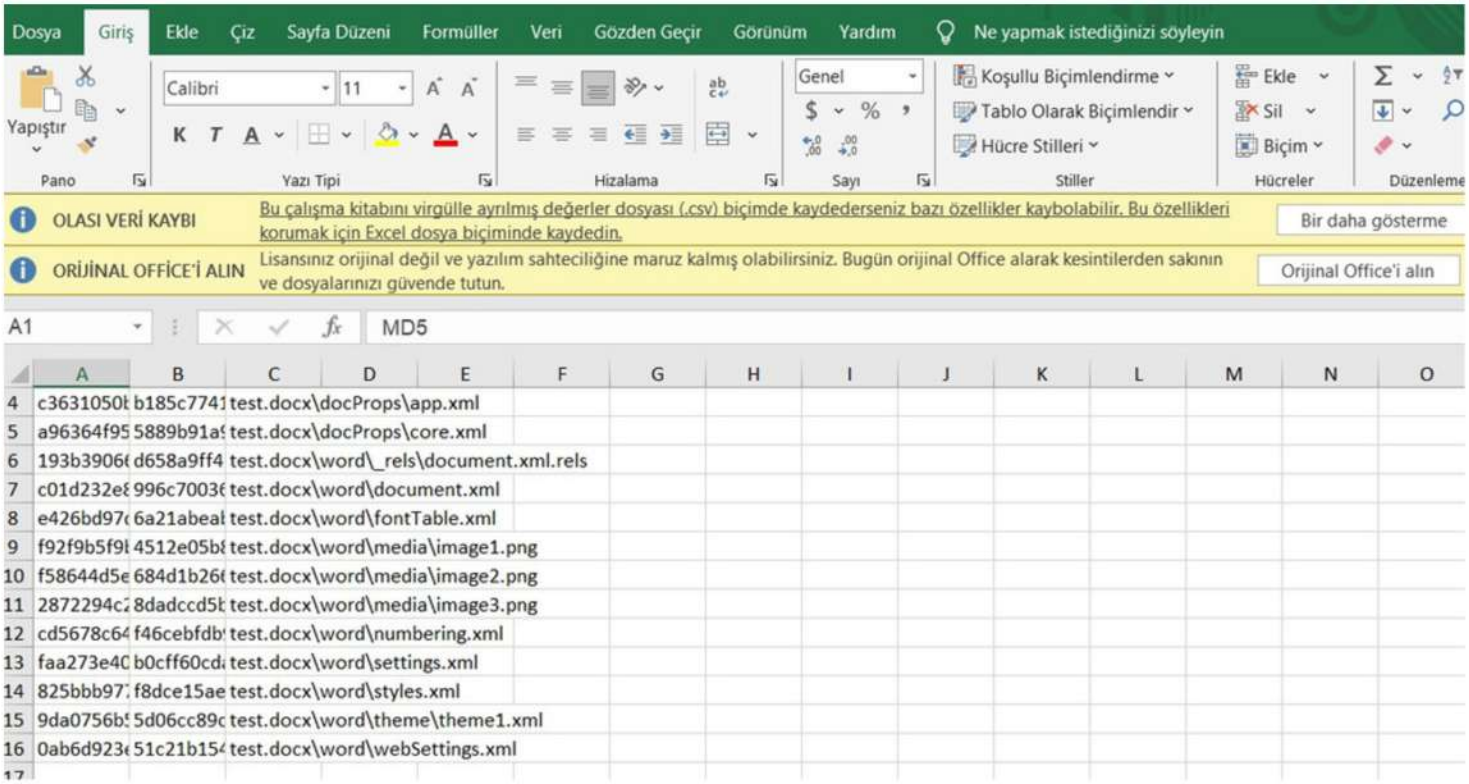
Menüden File > Add Evidence Item > ımage File > Next > Browse > Finish > Export File Hash List > Kaydet > Hash değeri .csv olarak kaydedildi





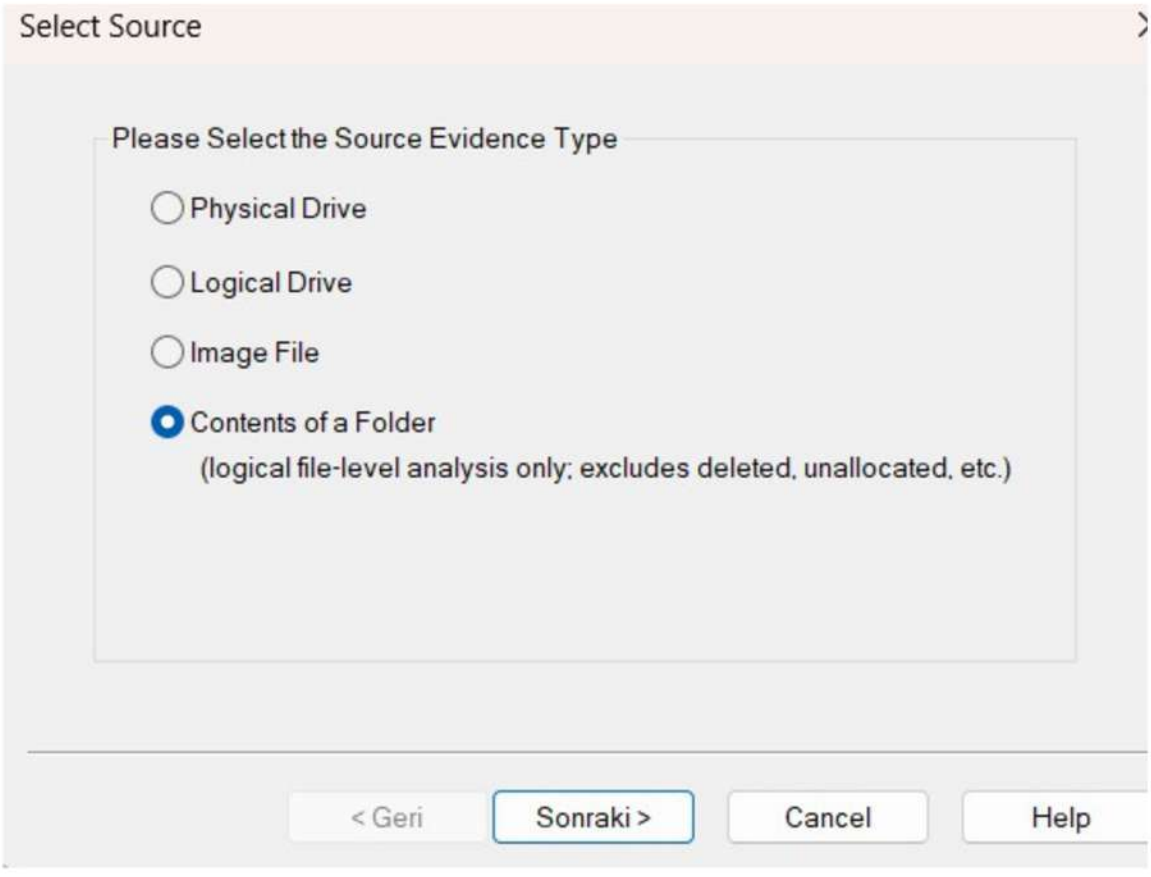

File View Mode Help
Add Evidence Item...
Add All Attached Devices
Image Mounting...
Remove Evidence Item
Remove All Evidence Items
Create Disk Image...
Export Disk Image...
Export Logical Image (AD1)...
Add to Custom Content Image (AD1)
Create Custom Content Image (AD1)...
Decrypt AD1 image...
Verify Drive/Image...
Capture Memory...
Obtain Protected Files...
Detect EFS Encryption
Export Files...
Export File Hash List...
Export Directory Listing...
Exit
List
Size
Type
Date Modified

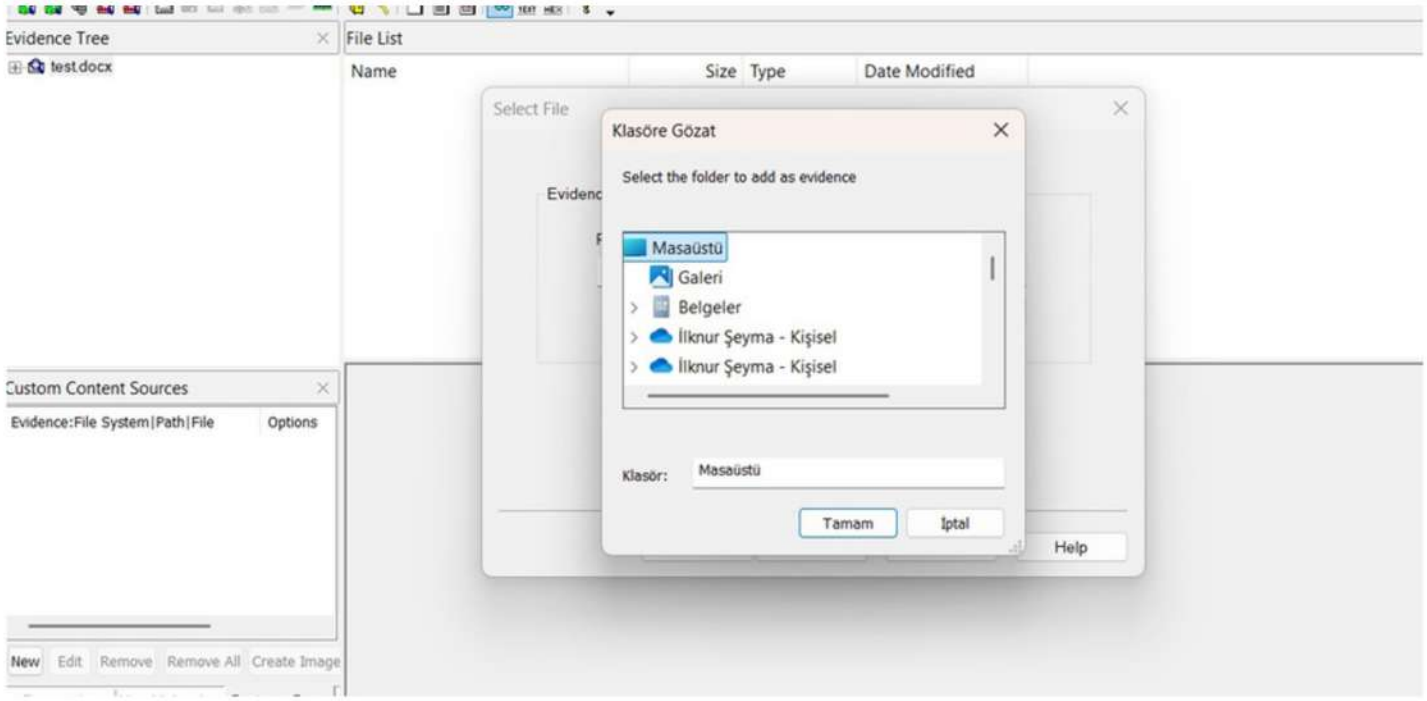

Düzenle
Yeni klasör
Masaüstü
Resimler
İlknur Şeyma - K
Masaüstü
İndirilenler
_20210316_ArabaKayit
CTF
AhrsApp
DERS NOTLARIM
Ajanda-Yaz-l-m--master
ESKİ klasör
akıllı trafik teknofest
felsefe
c# bitmiş proje
FGT_VM64-v7.0.10.M-build0450-FORTINET.out.ovf (2)
Dosya adı:
flaştesthash
Kayıt türü:
Comma Separated Values (*.csv)
Klasörleri Gizle
Kaydet
İptal

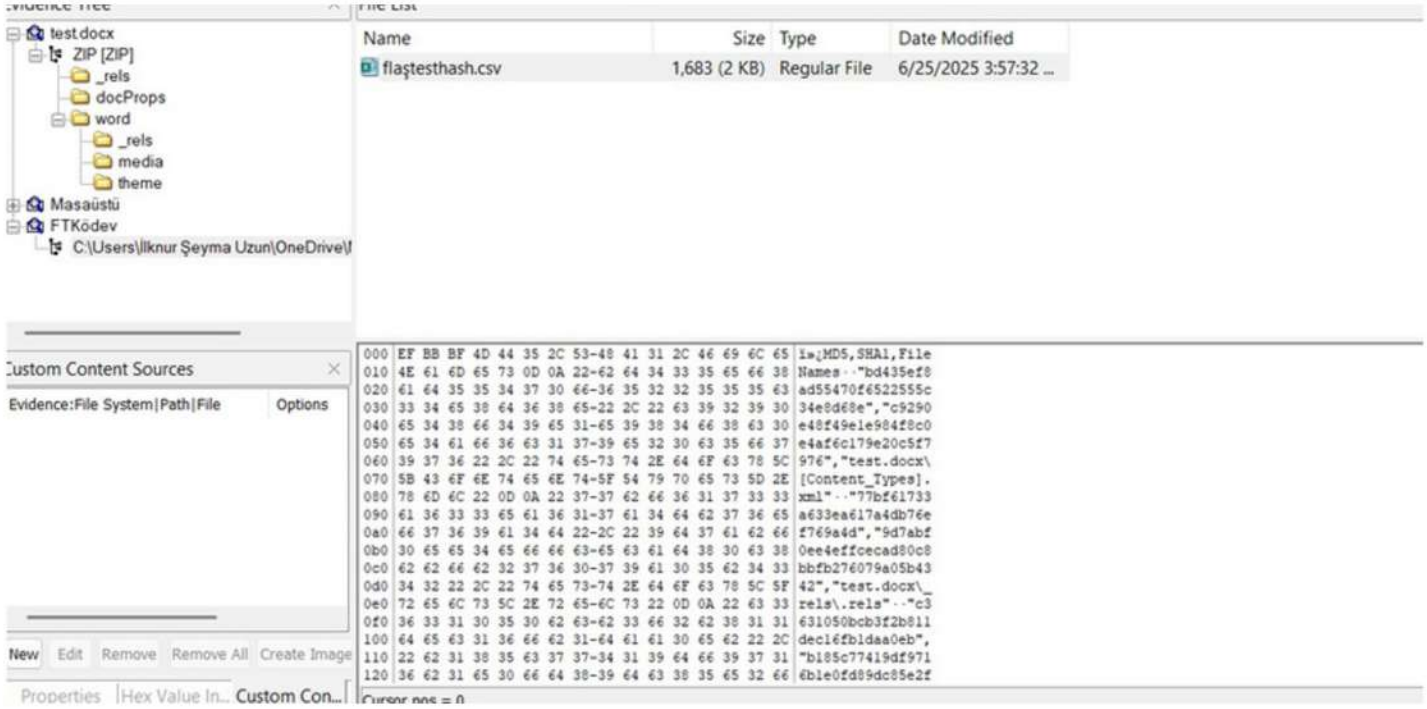


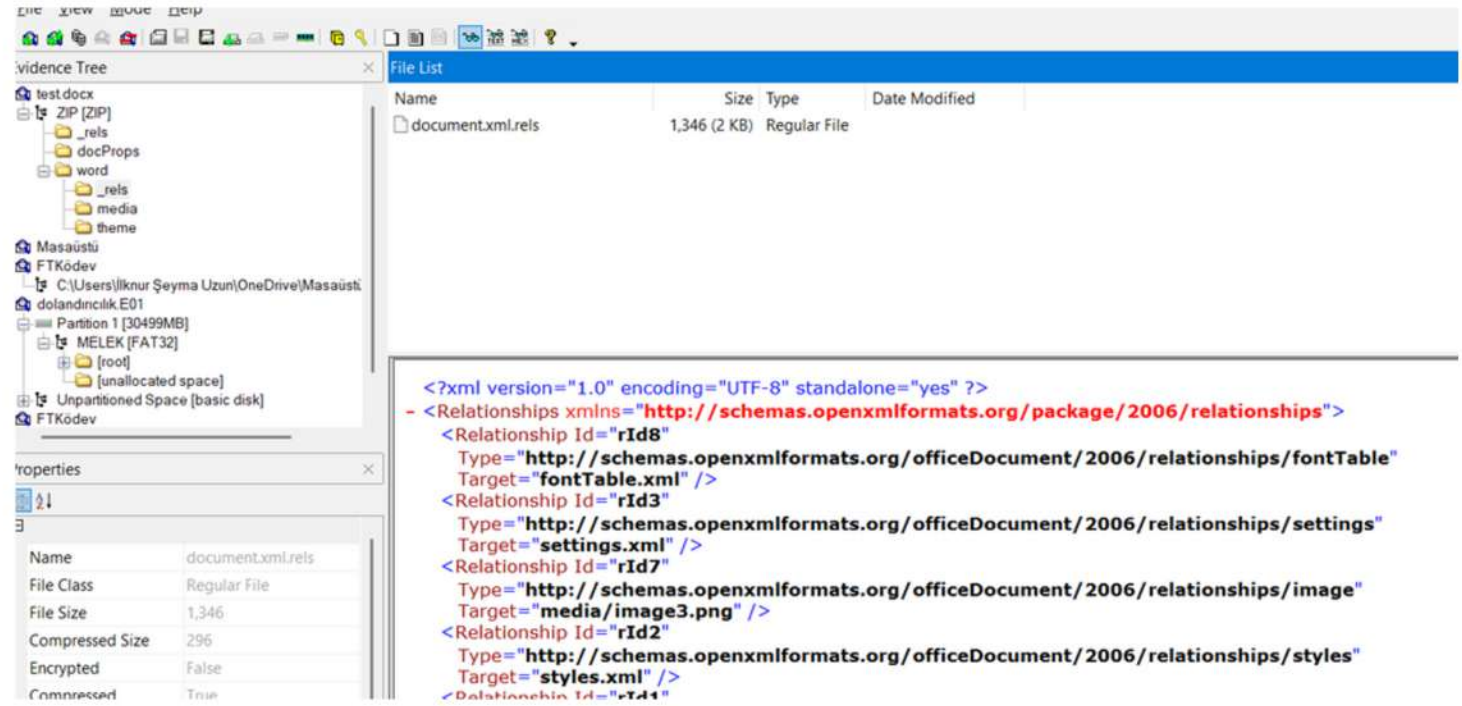
Masaüstündeki Klasör İçeriğinden Hash Alma

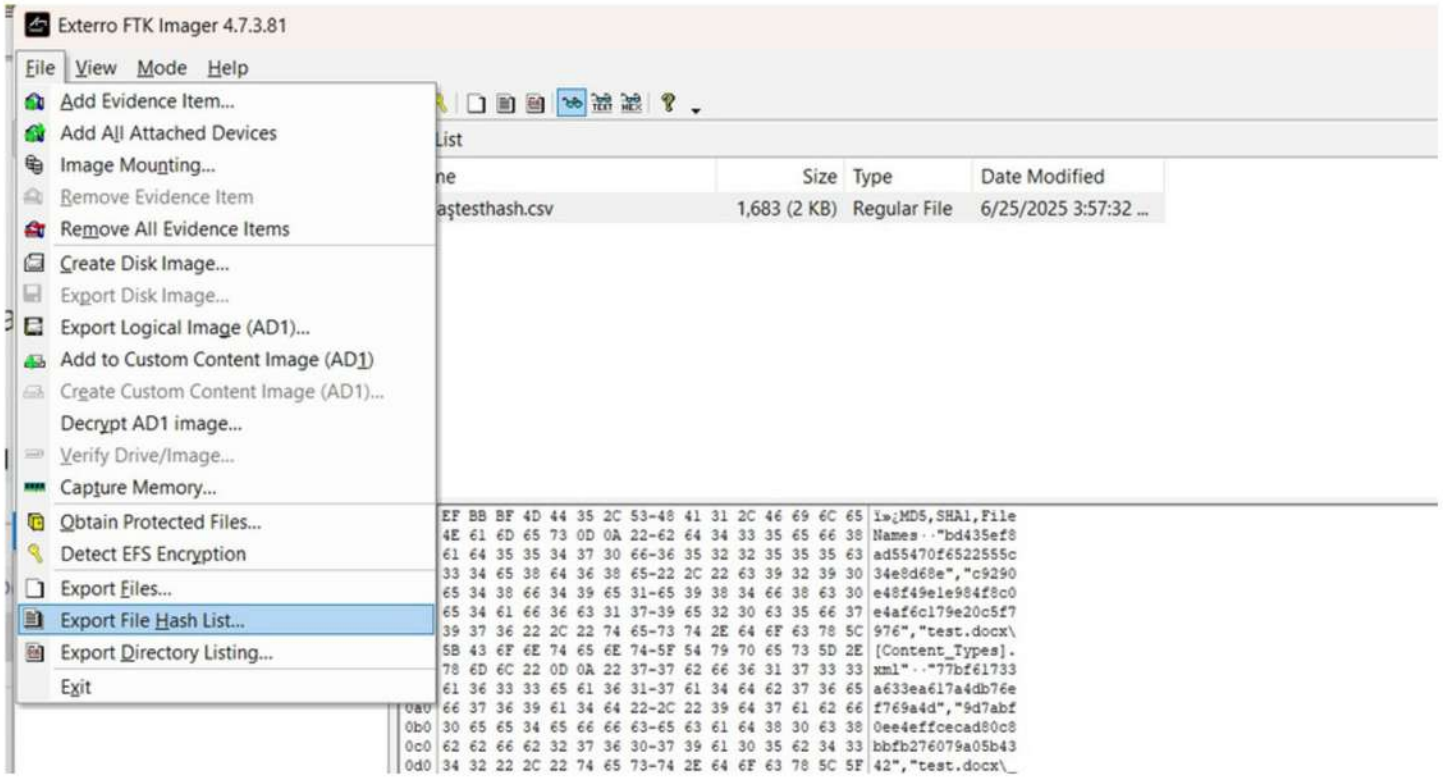
Contents of a Folder > Next >Browse > Masaüstü klasörünü > Finish> tree view (içerikte zıp, rels, docprops, word) > Export File Hash List (dosya adı vererek hash listesini kaydetti) > hash listesi .cvs olarak kaydedildi > properties (boyut dosya tam adı oluşturma zamanı vs.)

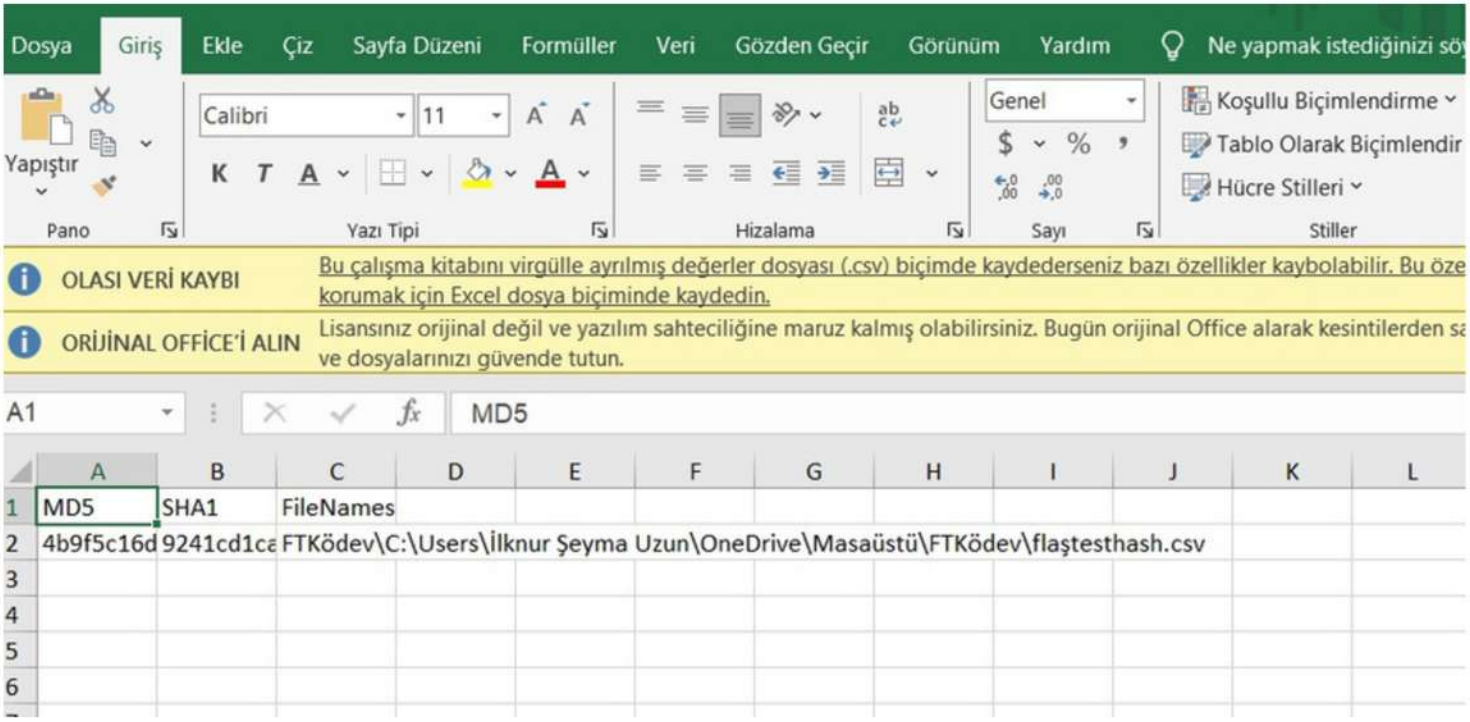


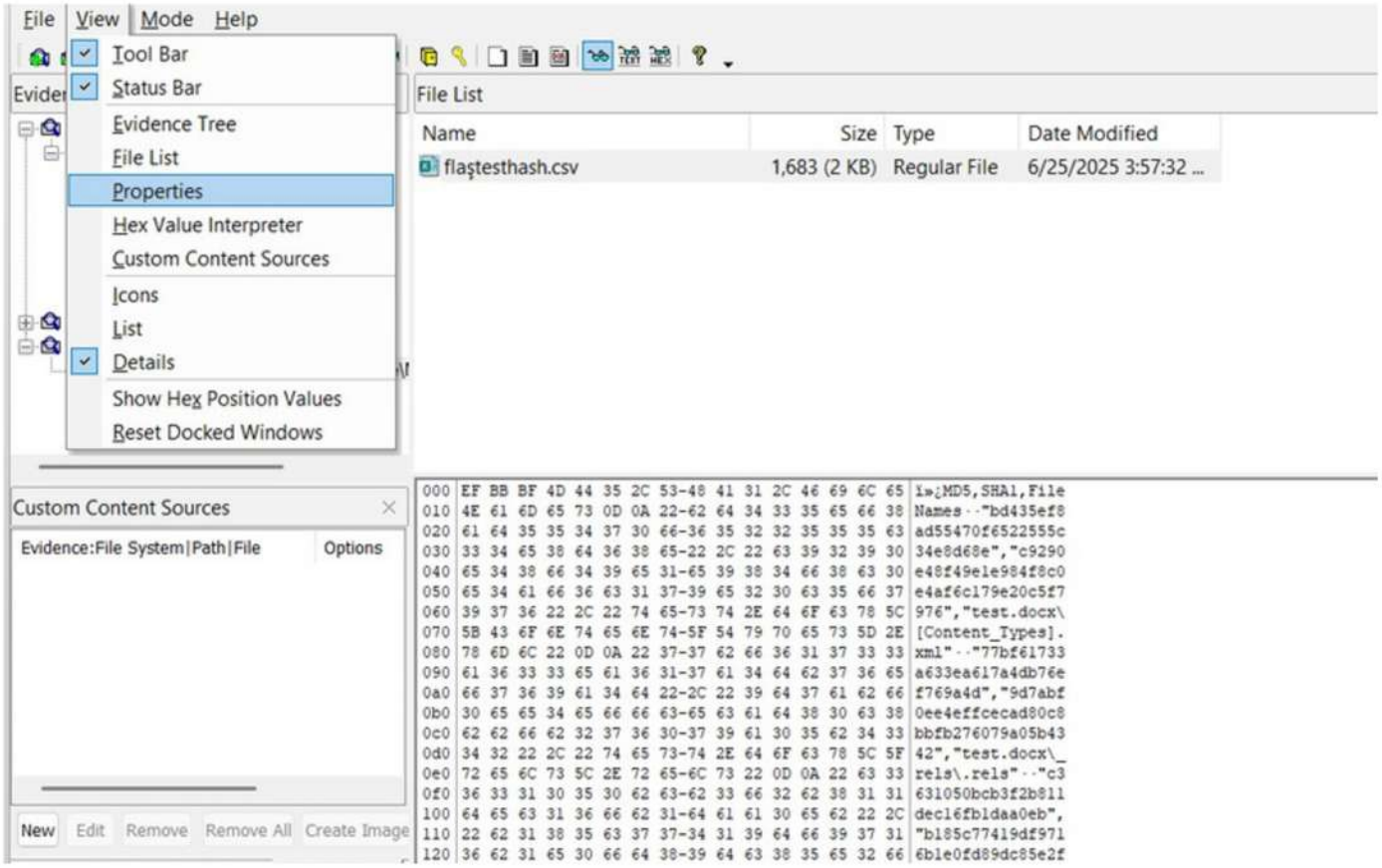

Evidence Tree
test.docx
File List
Name
Size
Type
Date Modified
Select File
Klasöre Gözat
Select the folder to add as evidence
Masaüstü
Galeri
Belgeler
İlknur Şeyma - Kişisel
İlknur Şeyma - Kişisel
Klasör:
Masaüstü
Tamam
İptal
Help
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image

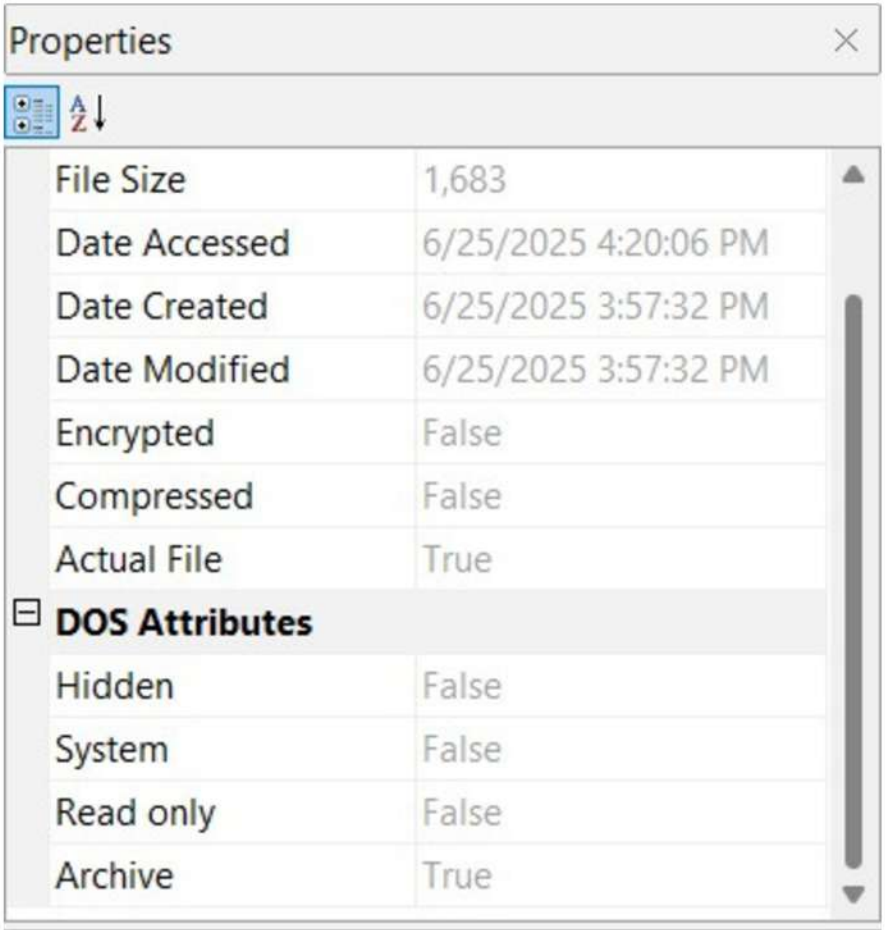

test.docx
ZIP [ZIP]
_rels
docProps
word
_rels
media
theme
Masaüstü
FTKödev
C:\Users\İlknur Şeyma Uzun\OneDrive\
Name
Size
Type
Date Modified
flaştesthash.csv
1,683 (2 KB)
Regular File
6/25/2025 3:57:32 ...
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value In...
Custom Con...
000 EF BB BF 4D 44 35 2C 53-48 41 31 2C 46 69 6C 65 i»¿MD5,SHA1,File
010 4E 61 6D 65 73 0D 0A 22-62 64 34 33 35 65 66 38 Names··"bd435ef8
020 61 64 35 35 34 37 30 66-36 35 32 32 35 35 35 63 ad55470f6522555c
030 33 34 65 38 64 36 38 65-22 2C 22 63 39 32 39 30 34e8d68e","c9290
040 65 34 38 66 34 39 65 31-65 39 38 34 66 38 63 30 e48f49e1e984f8c0
050 65 34 61 66 36 63 31 37-39 65 32 30 63 35 66 37 e4af6c179e20c5f7
060 39 37 36 22 2C 22 74 65-73 74 2E 64 6F 63 78 5C 976","test.docx\
070 5B 43 6F 6E 74 65 6E 74-5F 54 79 70 65 73 5D 2E [Content_Types].
080 78 6D 6C 22 0D 0A 22 37-37 62 66 36 31 37 33 33 xml"··"77bf61733
090 61 36 33 33 65 61 36 31-37 61 34 64 62 37 36 65 a633ea617a4db76e
0a0 66 37 36 39 61 34 64 22-2C 22 39 64 37 61 62 66 f769a4d","9d7abf
0b0 30 65 65 34 65 66 66 63-65 63 61 64 38 30 63 38 0ee4effcecad80c8
0c0 62 62 66 62 32 37 36 30-37 39 61 30 35 62 34 33 bbfb276079a05b43
0d0 34 32 22 2C 22 74 65 73-74 2E 64 6F 63 78 5C 5F 42","test.docx_
0e0 72 65 6C 73 5C 2E 72 65-6C 73 22 0D 0A 22 63 33 rels\.rels"··"c3
0f0 36 33 31 30 35 30 62 63-62 33 66 32 62 38 31 31 631050bcb3f2b811
100 64 65 63 31 36 66 62 31-64 61 61 30 65 62 22 2C dec16fb1daa0eb",
110 22 62 31 38 35 63 37 37-34 31 39 64 66 39 37 31 "b185c77419df971
120 36 62 31 65 30 66 64 38-39 64 63 38 35 65 32 66 6b1e0fd89dc85e2f
Cursor pos = 0


Evidence Tree
test.docx
ZIP [ZIP]
_rels
docProps
word
_rels
media
theme
Masaüstü
FTKödev
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü
dolandırıcılık.E01
Partition 1 [30499MB]
MELEK [FAT32]
[root]
[unallocated space]
Unpartitioned Space [basic disk]
FTKödev
File List
Name
Size
Type
Date Modified
document.xml.rels
1,346 (2 KB)
Regular File
Properties
Name
document.xml.rels
File Class
Regular File
File Size
1,346
Compressed Size
296
Encrypted
False
Compressed
True
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
<Relationships xmlns="http://schemas.openxmlformats.org/package/2006/relationships">
<Relationship Id="rId8"
Type="http://schemas.openxmlformats.org/officeDocument/2006/relationships/fontTable"
Target="fontTable.xml" />
<Relationship Id="rId3"
Type="http://schemas.openxmlformats.org/officeDocument/2006/relationships/settings"
Target="settings.xml" />
<Relationship Id="rId7"
Type="http://schemas.openxmlformats.org/officeDocument/2006/relationships/image"
Target="media/image3.png" />
<Relationship Id="rId2"
Type="http://schemas.openxmlformats.org/officeDocument/2006/relationships/styles"
Target="styles.xml" />
<Relationship Id="rId1"


Exterro FTK Imager 4.7.3.81
File View Mode Help
Add Evidence Item...
Add All Attached Devices
Image Mounting...
Remove Evidence Item
Remove All Evidence Items
Create Disk Image...
Export Disk Image...
Export Logical Image (AD1)...
Add to Custom Content Image (AD1)
Create Custom Content Image (AD1)...
Decrypt AD1 image...
Verify Drive/Image...
Capture Memory...
Obtain Protected Files...
Detect EFS Encryption
Export Files...
Export File Hash List...
Export Directory Listing...
Exit
List
Size
Type
Date Modified
aştesthash.csv
1,683 (2 KB)
Regular File
6/25/2025 3:57:32 ...


Dosya
Giriş
Ekle
Çiz
Sayfa Düzeni
Formüller
Veri
Gözden Geçir
Görünüm
Yardım
Ne yapmak istediğinizi söy
Yapıştır
Calibri
11
Genel
Koşullu Biçimlendirme
Tablo Olarak Biçimlendir
Hücre Stilleri
Pano
Yazı Tipi
Hizalama
Sayı
Stiller
OLASI VERİ KAYBI
Bu çalışma kitabını virgülle ayrılmış değerler dosyası (.csv) biçimde kaydederseniz bazı özellikler kaybolabilir. Bu öze
korumak için Excel dosya biçiminde kaydedin.
ORİJİNAL OFFİCE'İ ALIN
Lisansınız orijinal değil ve yazılım sahteciliğine maruz kalmış olabilirsiniz. Bugün orijinal Office alarak kesintilerden sa
ve dosyalarınızı güvende tutun.
A1
MD5
MD5
SHA1
FileNames
4b9f5c16d
9241cd1ca
FTKödev\C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev\flaştesthash.csv

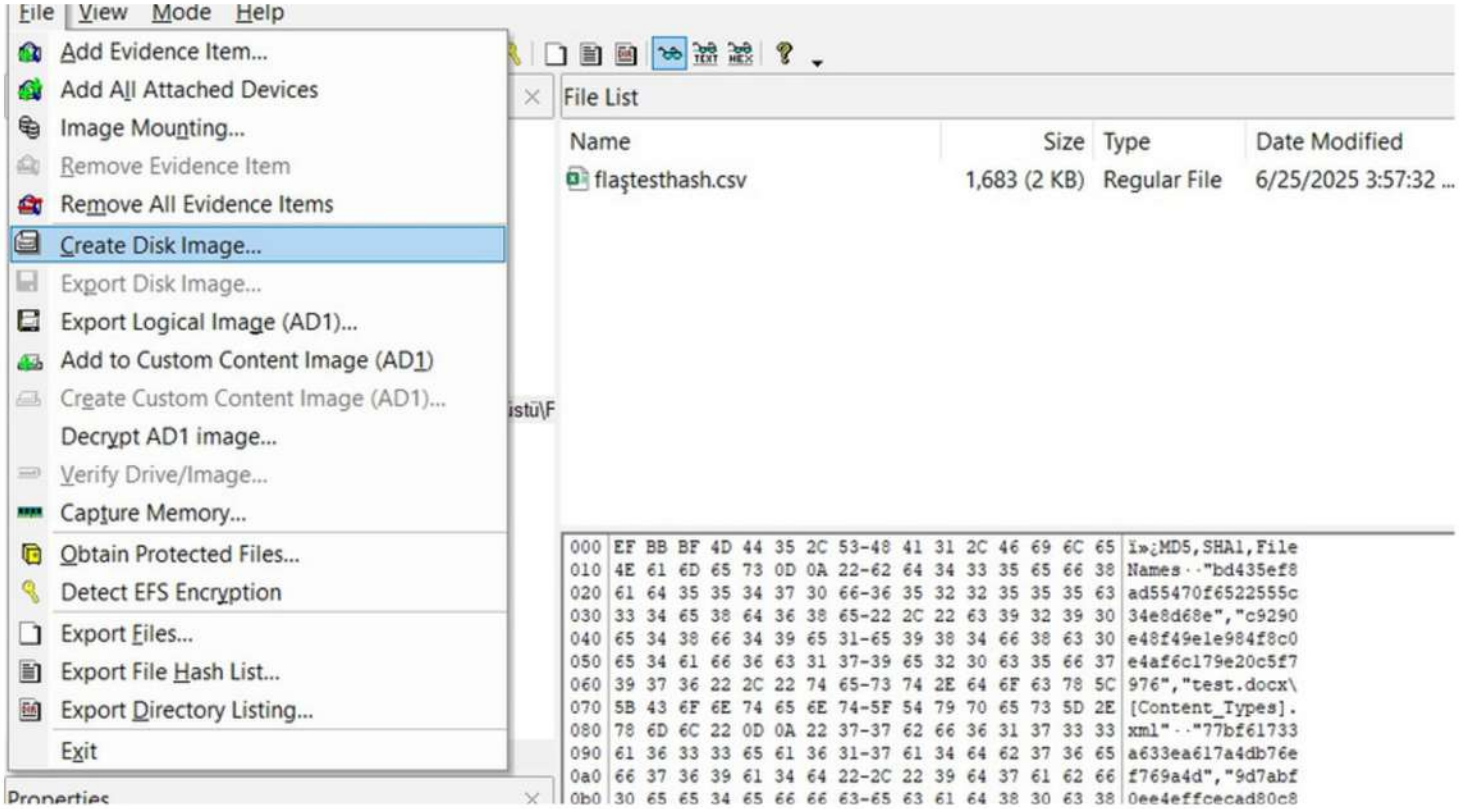

File View Mode Help
Tool Bar
Status Bar
Evidence Tree
File List
Properties
Hex Value Interpreter
Custom Content Sources
Icons
List
Details
Show Hex Position Values
Reset Docked Windows
File List
Name
Size
Type
Date Modified
flaştesthash.csv
1,683 (2 KB)
Regular File
6/25/2025 3:57:32 ...
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
000 EF BB BF 4D 44 35 2C 53-48 41 31 2C 46 69 6C 65 i»¿MD5,SHA1,File
010 4E 61 6D 65 73 0D 0A 22-62 64 34 33 35 65 66 38 Names··"bd435ef8
020 61 64 35 35 34 37 30 66-36 35 32 32 35 35 35 63 ad55470f6522555c
030 33 34 65 38 64 36 38 65-22 2C 22 63 39 32 39 30 34e8d68e","c9290
040 65 34 38 66 34 39 65 31-65 39 38 34 66 38 63 30 e48f49e1e984f8c0
050 65 34 61 66 36 63 31 37-39 65 32 30 63 35 66 37 e4af6c179e20c5f7
060 39 37 36 22 2C 22 74 65-73 74 2E 64 6F 63 78 5C 976","test.docx\
070 5B 43 6F 6E 74 65 6E 74-5F 54 79 70 65 73 5D 2E [Content_Types].
080 78 6D 6C 22 0D 0A 22 37-37 62 66 36 31 37 33 33 xml"··"77bf61733
090 61 36 33 33 65 61 36 31-37 61 34 64 62 37 36 65 a633ea617a4db76e
0a0 66 37 36 39 61 34 64 22-2C 22 39 64 37 61 62 66 f769a4d","9d7abf
0b0 30 65 65 34 65 66 66 63-65 63 61 64 38 30 63 38 0ee4effcecad80c8
0c0 62 62 66 62 32 37 36 30-37 39 61 30 35 62 34 33 bbfb276079a05b43
0d0 34 32 22 2C 22 74 65 73-74 2E 64 6F 63 78 5C 5F 42","test.docx_
0e0 72 65 6C 73 5C 2E 72 65-6C 73 22 0D 0A 22 63 33 rels\.rels"··"c3
0f0 36 33 31 30 35 30 62 63-62 33 66 32 62 38 31 31 631050bcb3f2b811
100 64 65 63 31 36 66 62 31-64 61 61 30 65 62 22 2C dec16fb1daa0eb",
110 22 62 31 38 35 63 37 37-34 31 39 64 66 39 37 31 "b185c77419df971
120 36 62 31 65 30 66 64 38-39 64 63 38 35 65 32 66 6b1e0fd89dc85e2f

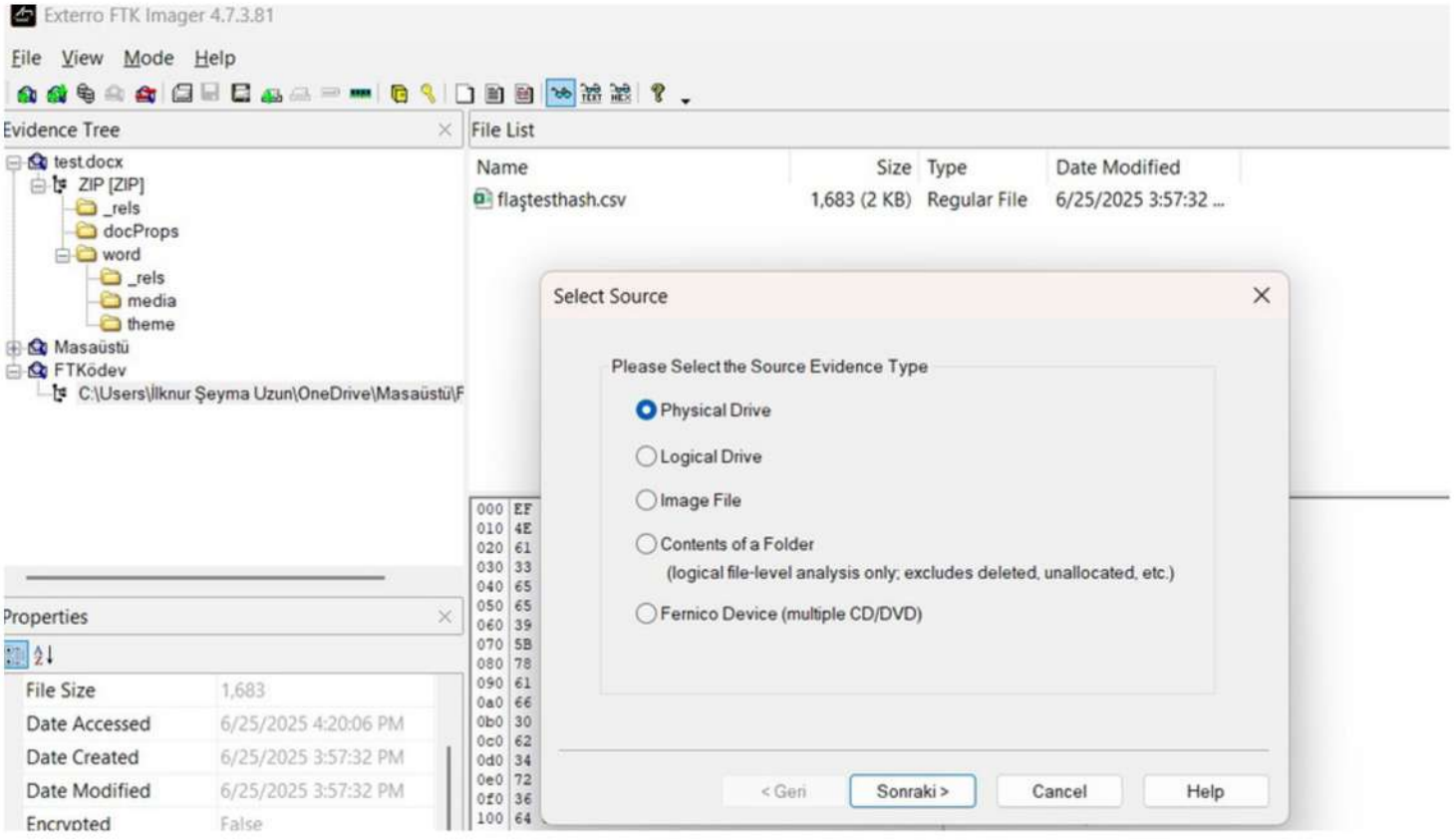

Properties
File Size
1,683
Date Accessed
6/25/2025 4:20:06 PM
Date Created
6/25/2025 3:57:32 PM
Date Modified
6/25/2025 3:57:32 PM
Encrypted
False
Compressed
False
Actual File
True
DOS Attributes
Hidden
False
System
False
Read only
False
Archive
True

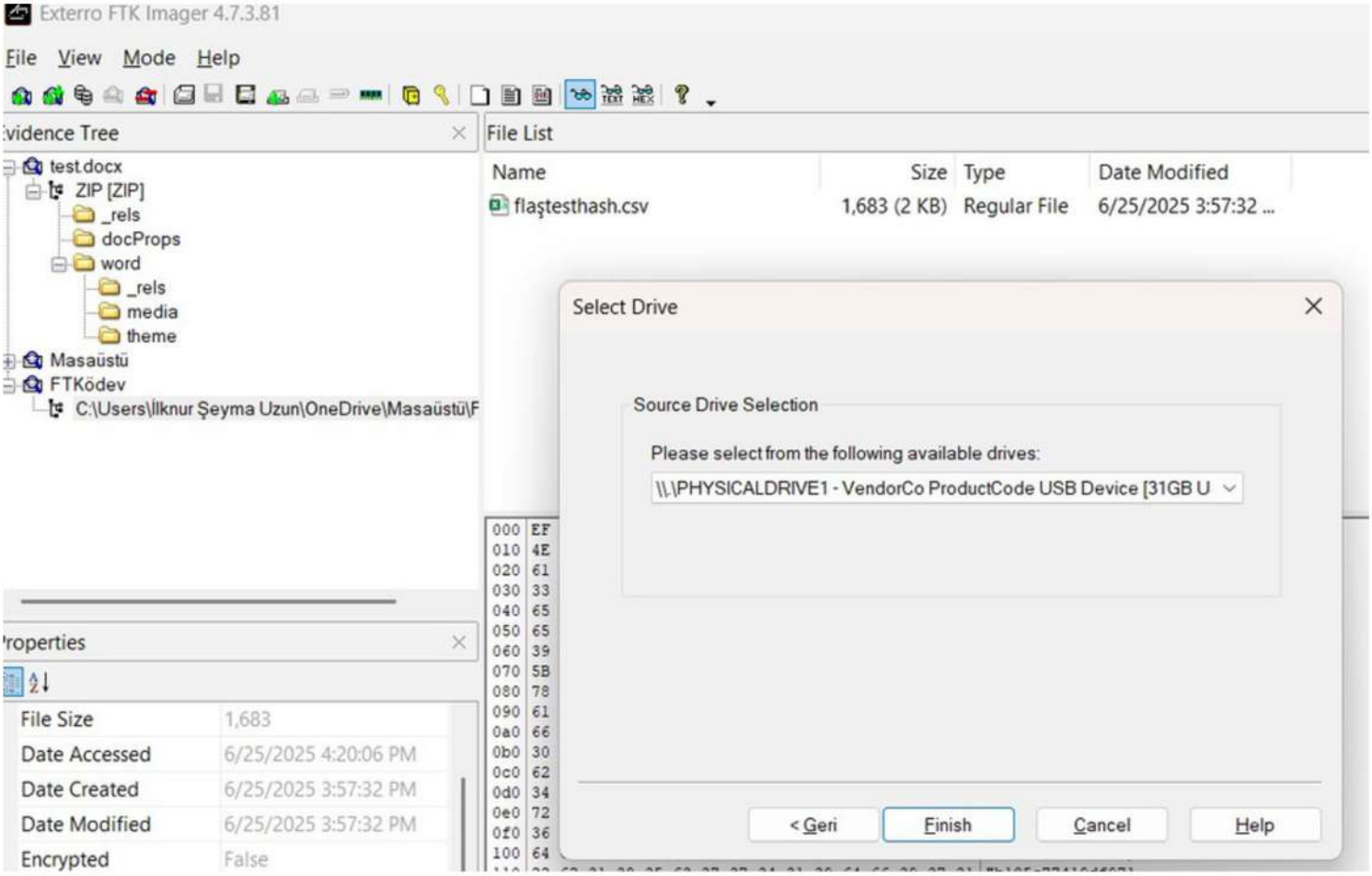
USB Bellek Imaj Alma

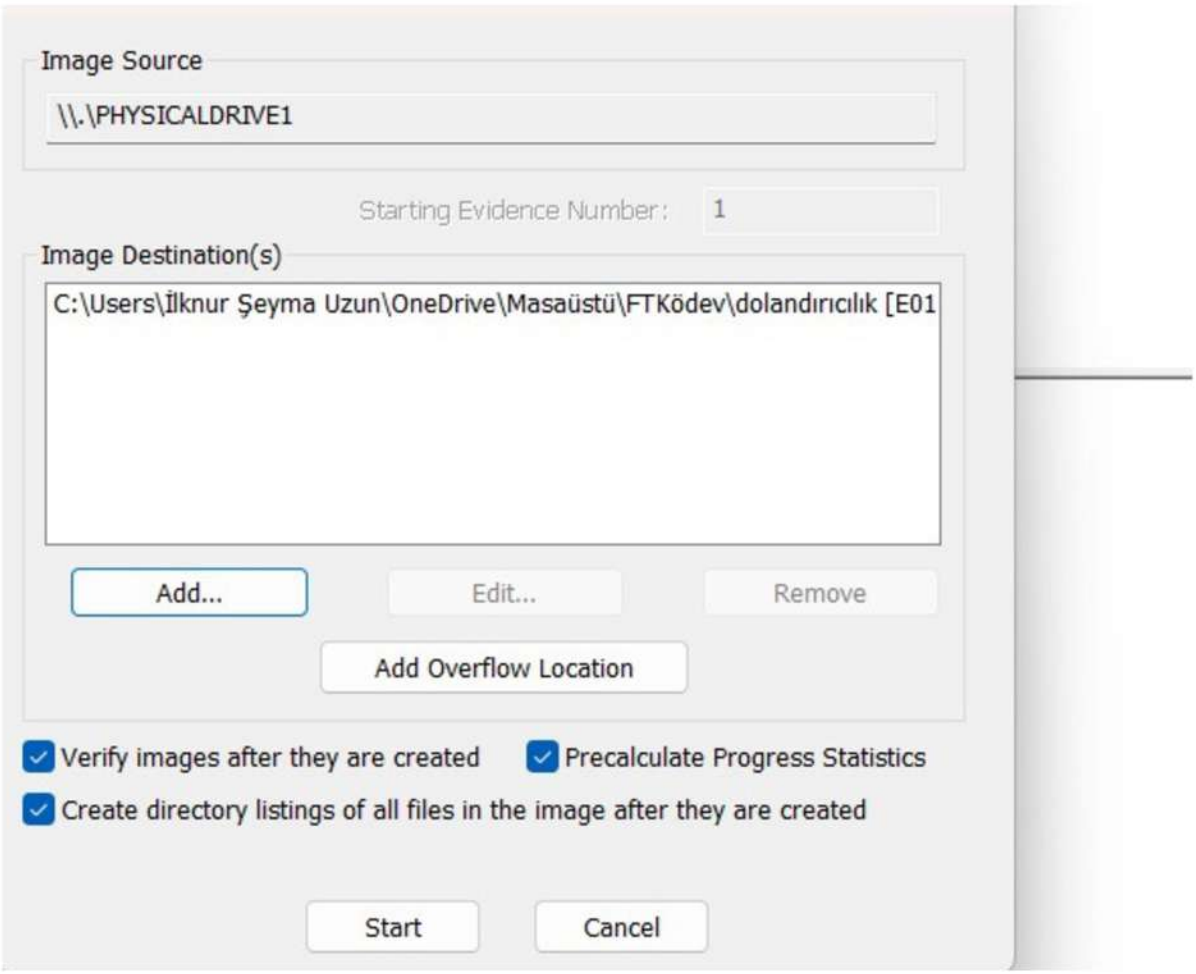
- **İmaj bir diskin birebir dijital kopyasıdır. Orijinal diske zarar vermemek için imaj alınır. İnceleme bu kopya üzerinde yapılır.**

File > Create Disk Image > Physical Drive > E01 > Evidence Item Information > Select Image Destination

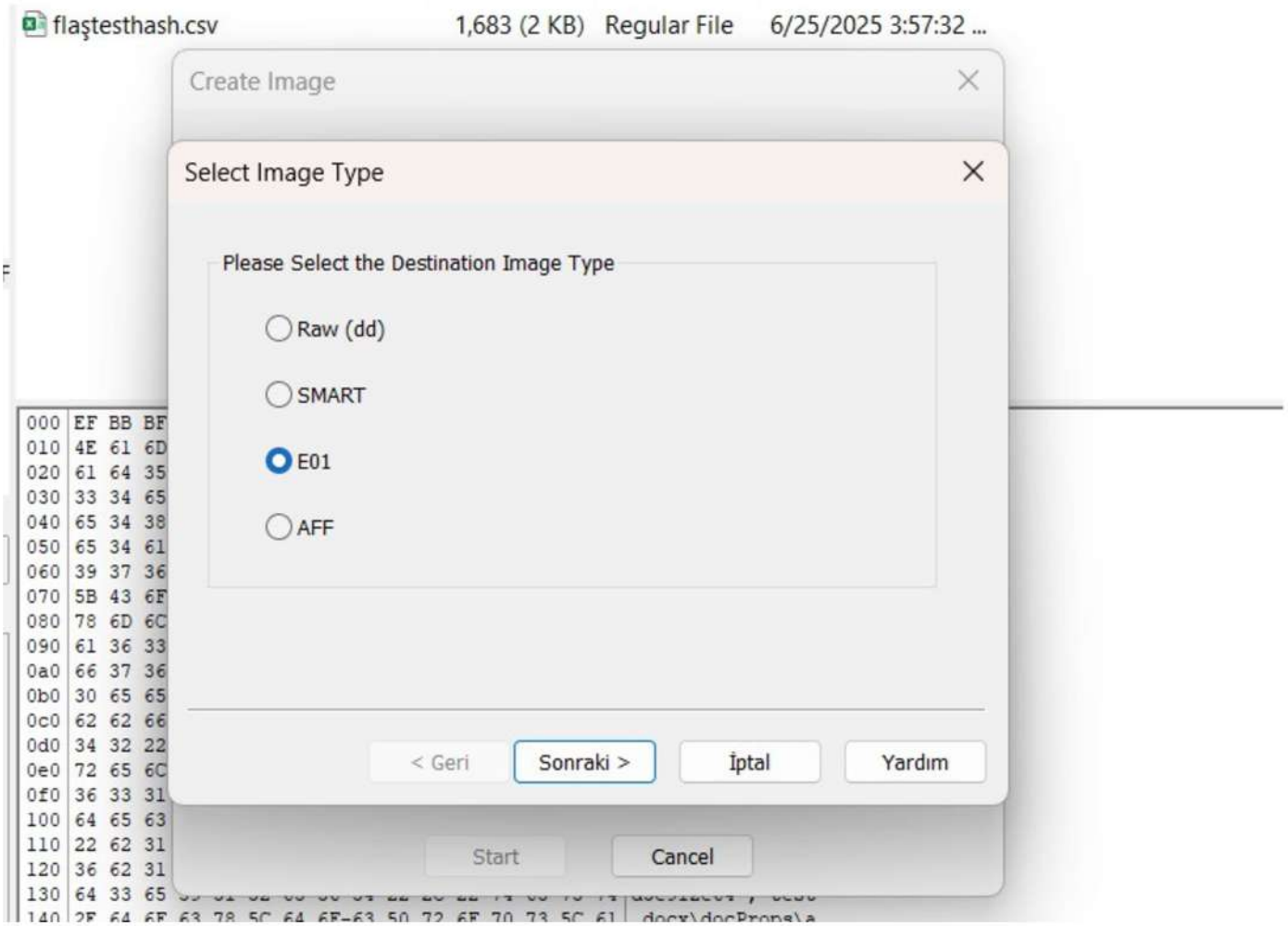


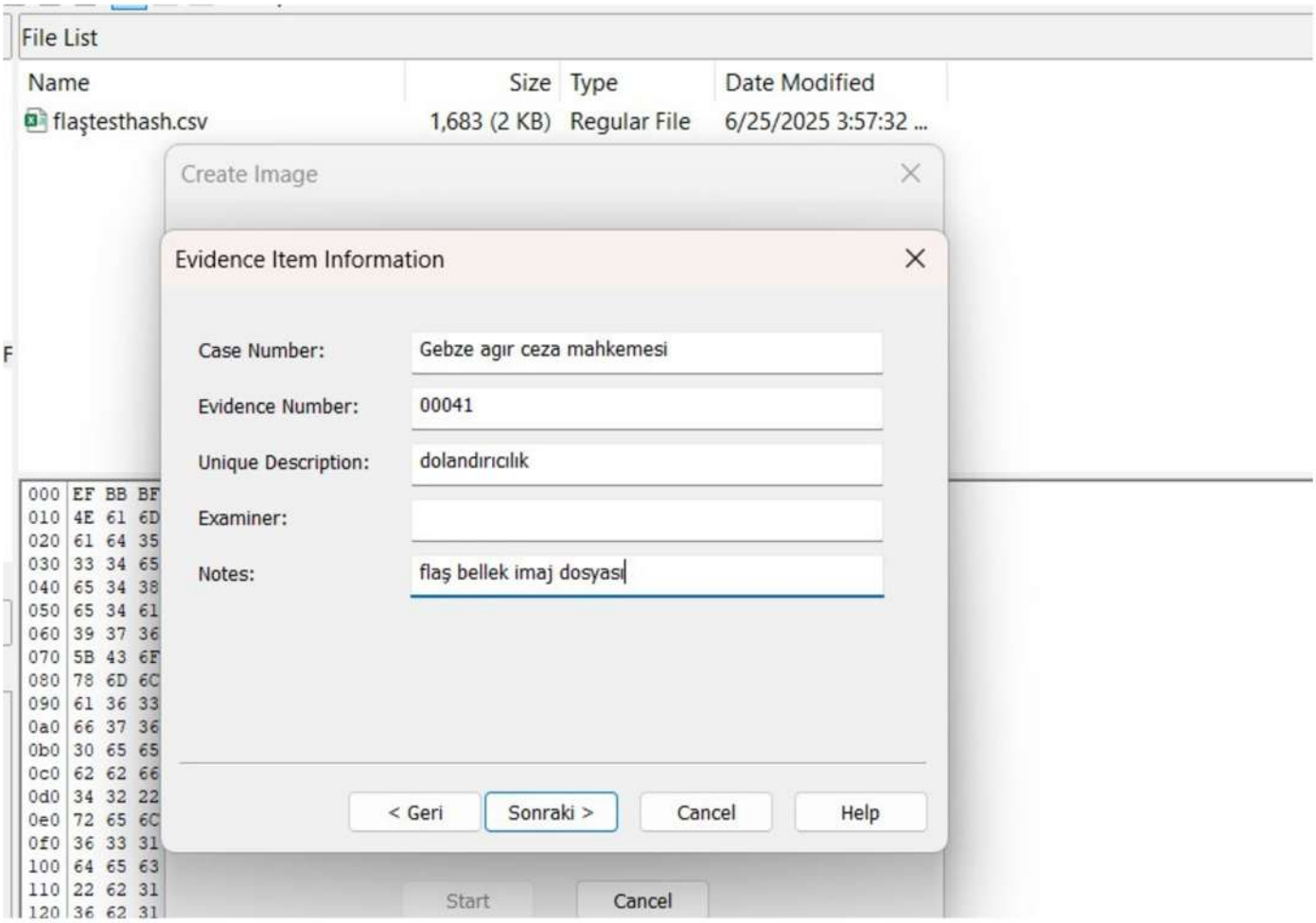


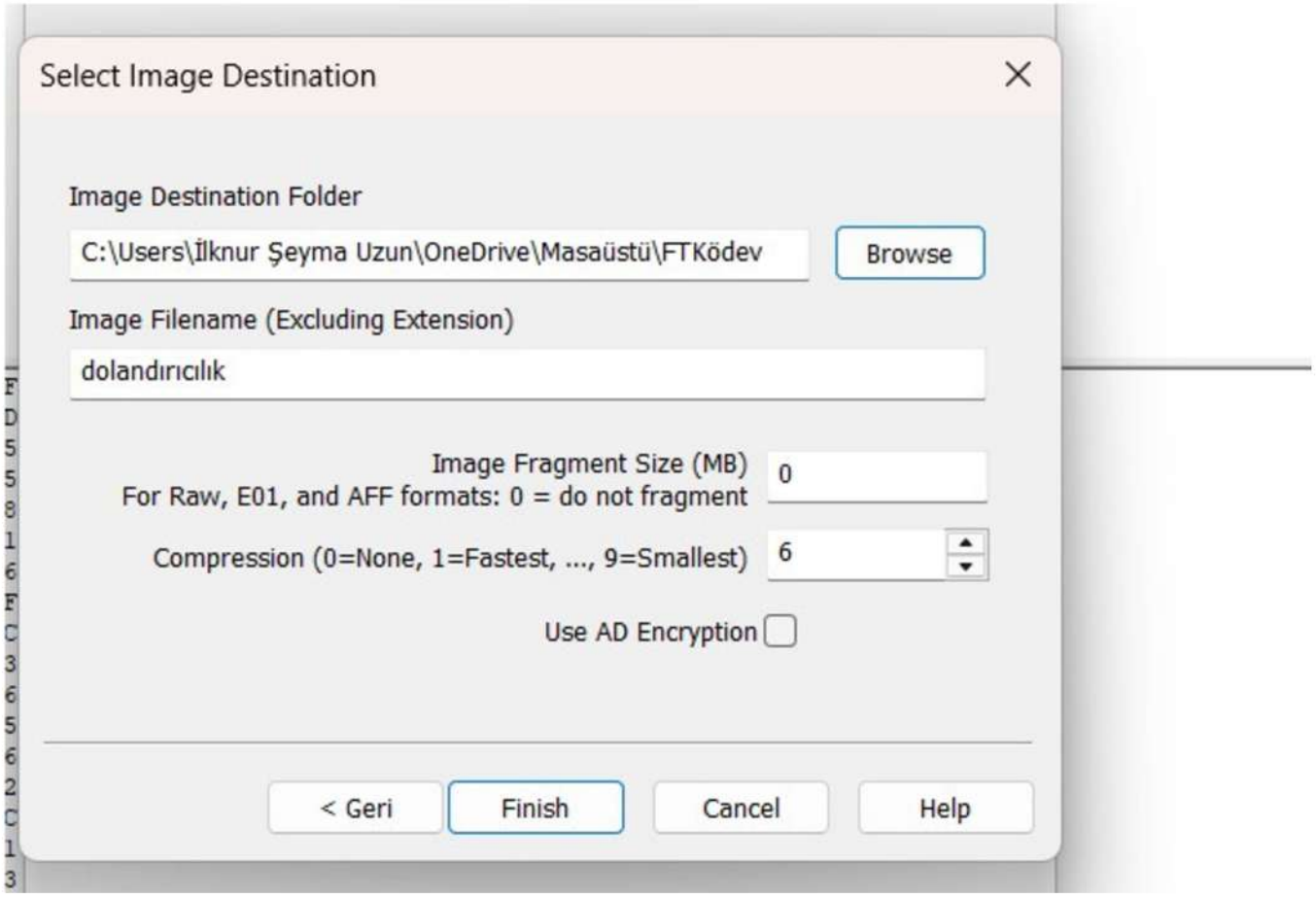

Exterro FTK Imager 4.7.3.81
File View Mode Help
Evidence Tree
test.docx
ZIP [ZIP]
_rels
docProps
word
_rels
media
theme
Masaüstü
FTKödev
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\F
File List
Name
Size
Type
Date Modified
flaştesthash.csv
1,683 (2 KB)
Regular File
6/25/2025 3:57:32 ...
Select Drive
Source Drive Selection
Please select from the following available drives:
\\.\PHYSICALDRIVE1 - VendorCo ProductCode USB Device [31GB U
< Geri
Finish
Cancel
Help
Properties
File Size
1,683
Date Accessed
6/25/2025 4:20:06 PM
Date Created
6/25/2025 3:57:32 PM
Date Modified
6/25/2025 3:57:32 PM
Encrypted
False

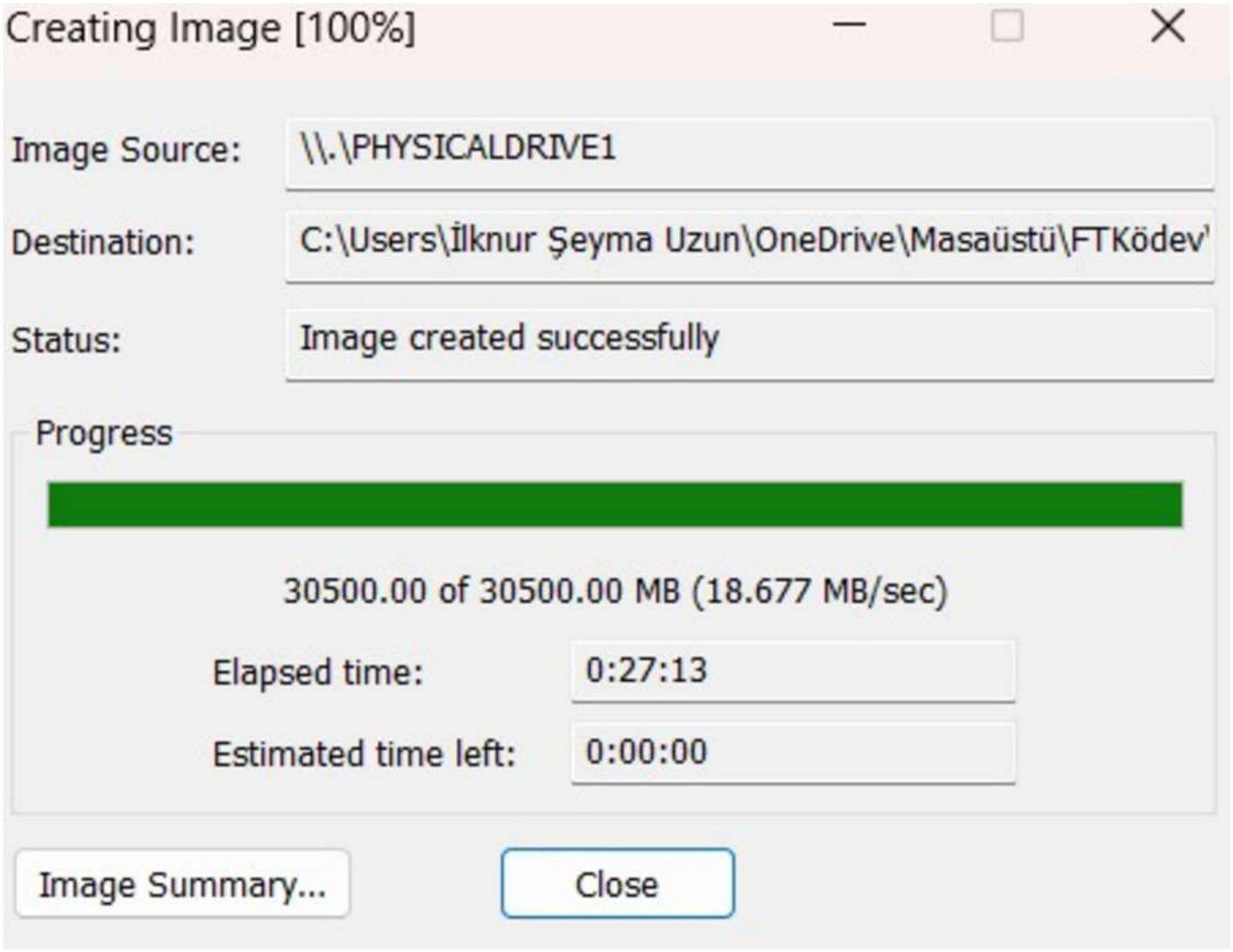

Image Source
\\.\PHYSICALDRIVE1
Starting Evidence Number:
1
Image Destination(s)
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev\dolandırıcılık [E01
Add...
Edit...
Remove
Add Overflow Location
Verify images after they are created
Precalculate Progress Statistics
Create directory listings of all files in the image after they are created
Start
Cancel

E01 dosyayı sıkıştırarak indirir 1 az sıkışmış 9 en fazla sıkışmış dosyadır, ortalama olarak 6-7 de indirmek daha iyidir.

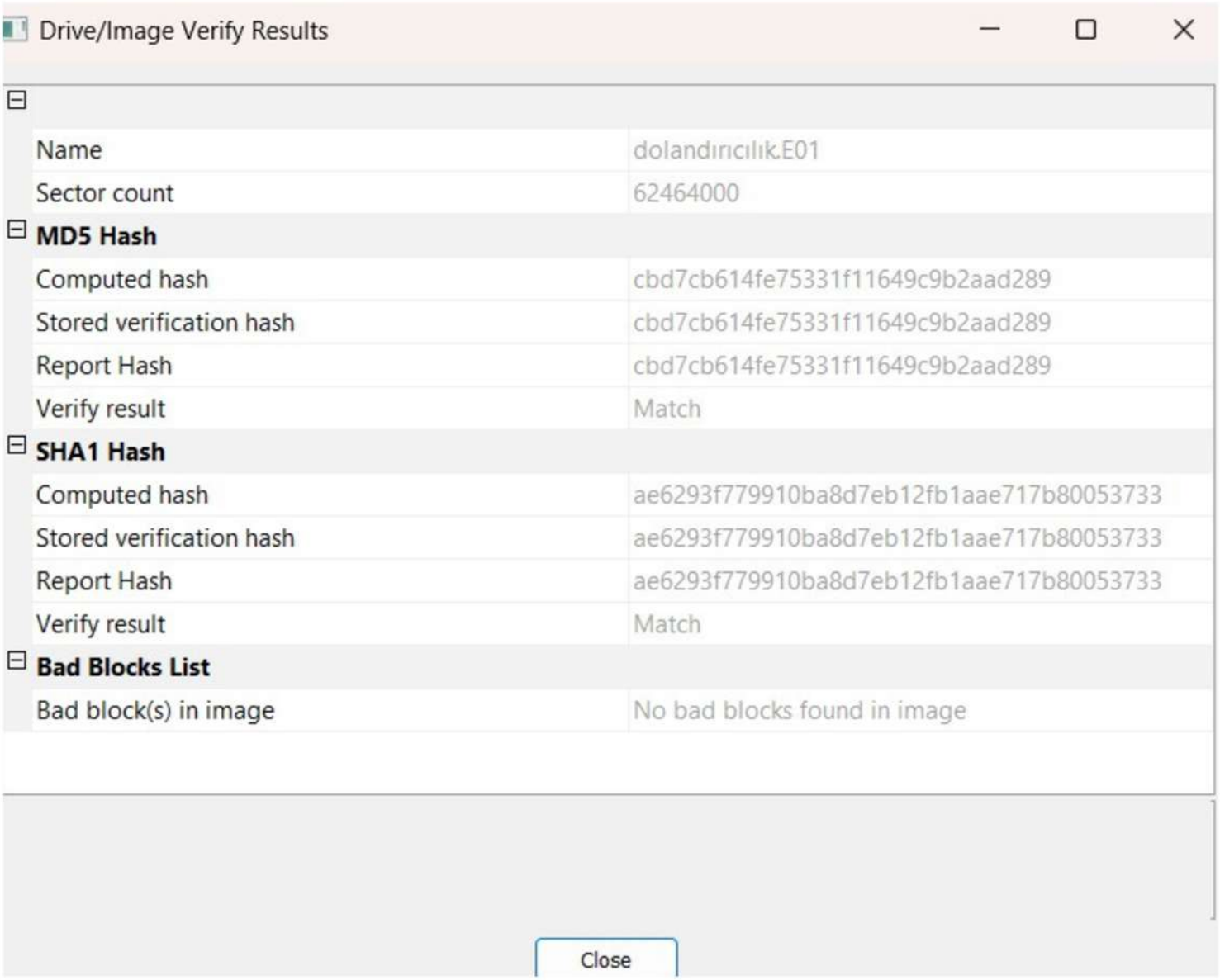




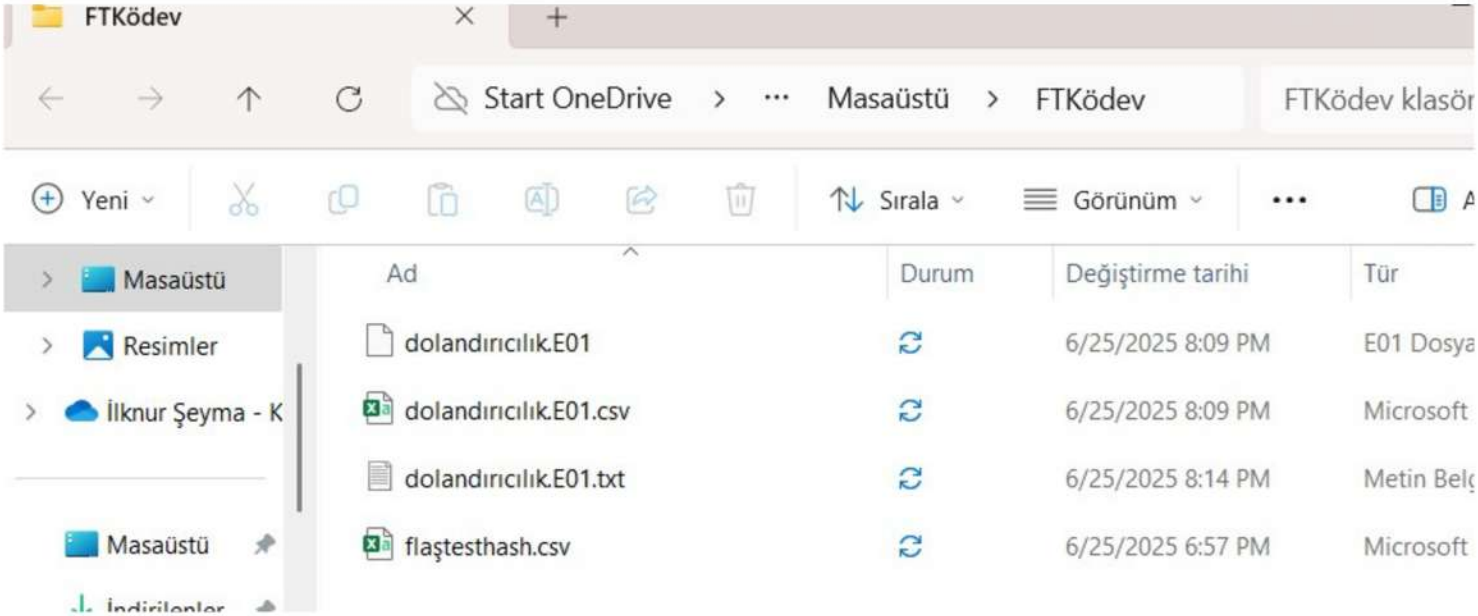

Select Image Destination
Image Destination Folder
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev
Browse
Image Filename (Excluding Extension)
dolandırıcılık
Image Fragment Size (MB)
For Raw, E01, and AFF formats: 0 = do not fragment
0
Compression (0=None, 1=Fastest, ..., 9=Smallest)
6
Use AD Encryption
< Geri
Finish
Cancel
Help


Creating Image [100%]
Image Source:
\\.\PHYSICALDRIVE1
Destination:
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev'
Status:
Image created successfully
Progress
30500.00 of 30500.00 MB (18.677 MB/sec)
Elapsed time:
0:27:13
Estimated time left:
0:00:00
Image Summary...
Close

Hash değeri ve işlemin başarılı olup olmadığını gösterir



E01 → İmaj dosyası
.txt → Log bilgisi
.csv → Dosya listesi ve hash bilgileri



```
Unique description: dolandırıcılık
Examiner:
Notes: flaş bellek imaj dosyası

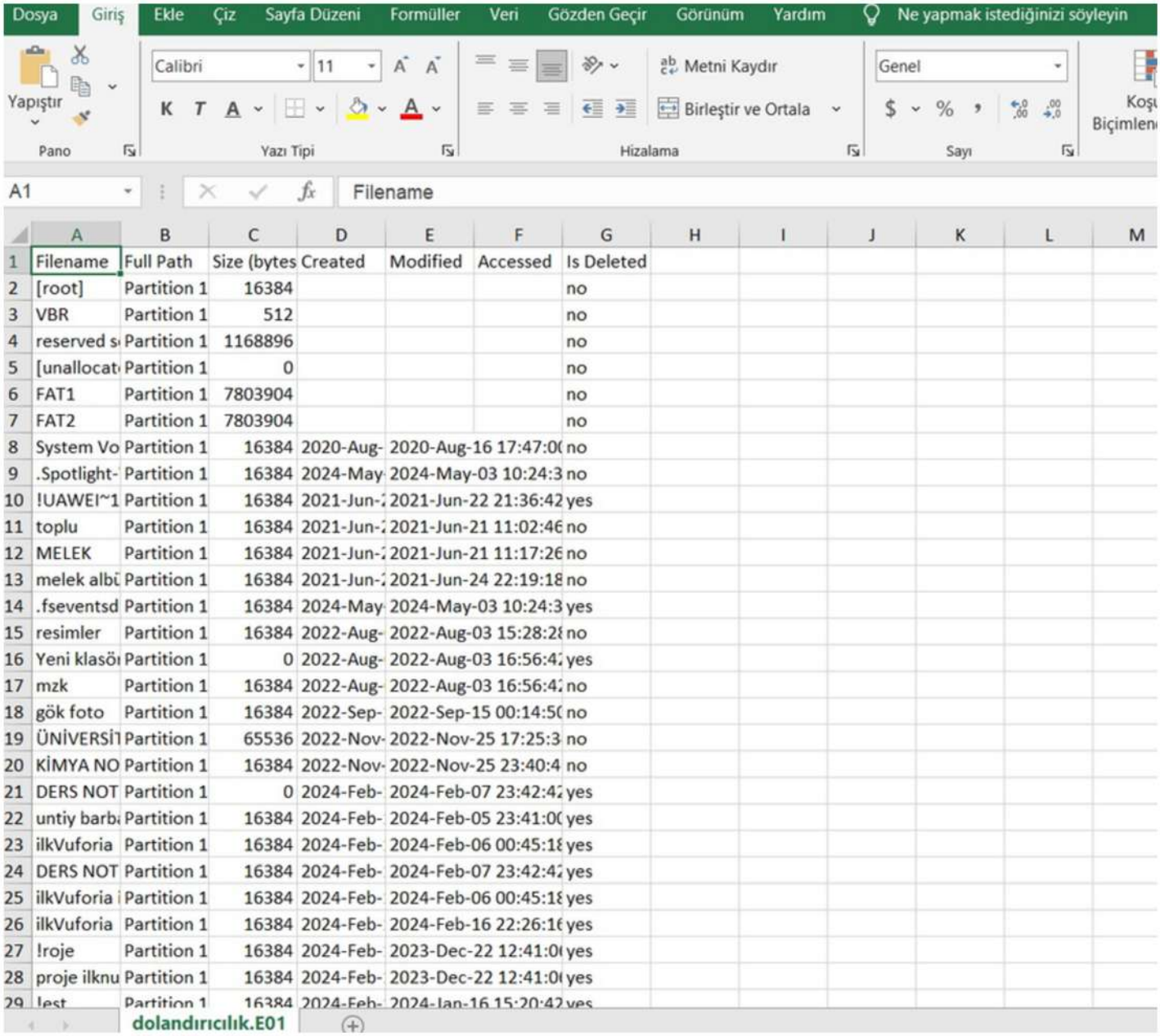
---------------------------------------------------------------

Information for C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev\dolandırıcılık:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 3,888
 Tracks per Cylinder: 255
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 62,464,000
[Physical Drive Information]
 Drive Model: VendorCo ProductCode USB Device
 Drive Serial Number: 2999691240859133357
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 30500 MB
 Sector count:    62464000
[Computed Hashes]
 MD5 checksum:    cbd7cb614fe75331f11649c9b2aad289
 SHA1 checksum:   ae6293f779910ba8d7eb12fb1aae717b80053733

Image Information:
 Acquisition started:   Wed Jun 25 19:42:30 2025
 Acquisition finished:  Wed Jun 25 20:09:43 2025
 Segment list:
  C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev\dolandırıcılık.E01
 COMPUTED HASH :  cbd7cb614fe75331f11649c9b2aad289
 COMPUTED HASH :  ae6293f779910ba8d7eb12fb1aae717b80053733

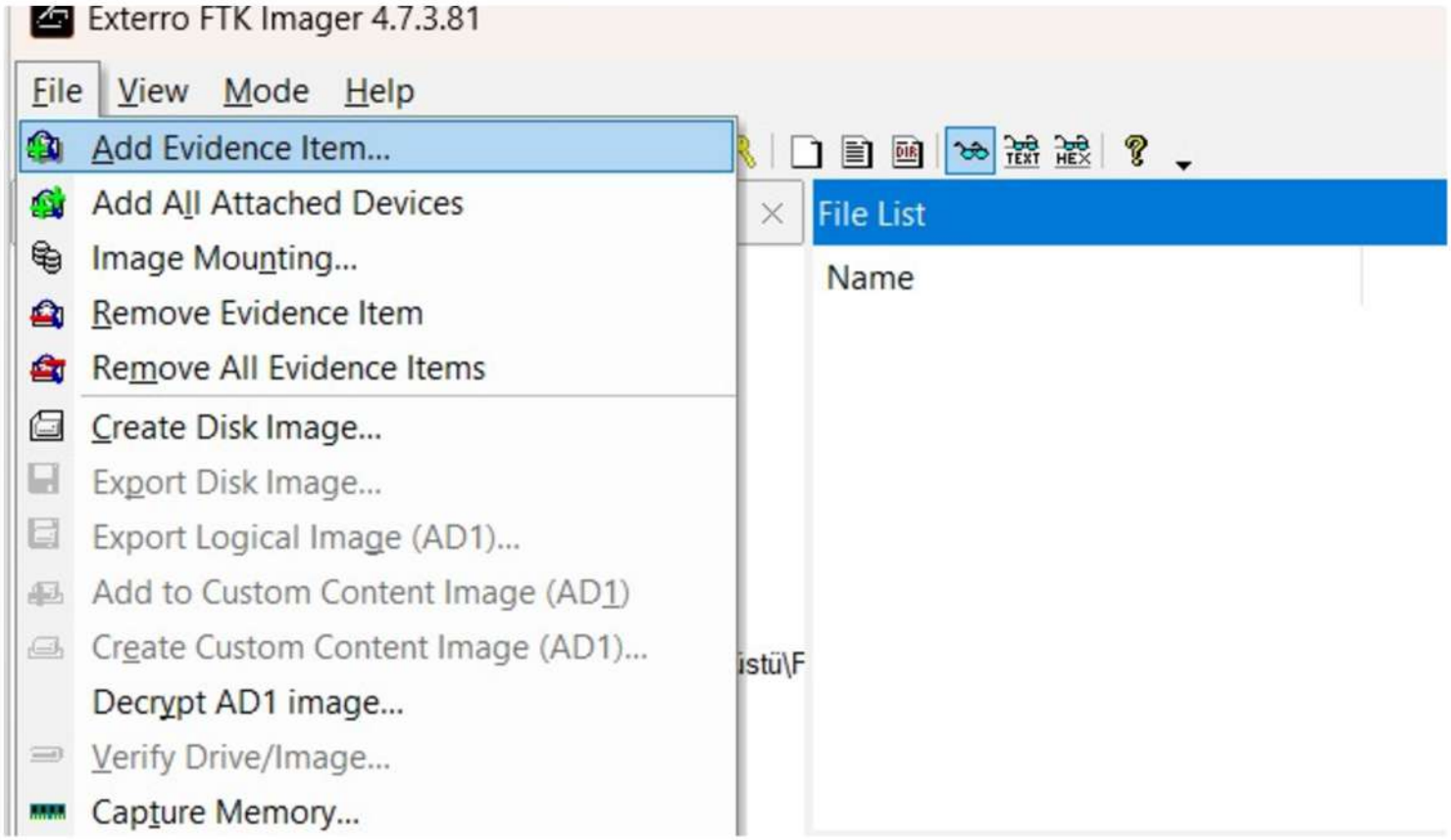
Image Verification Results:
 Verification started:  Wed Jun 25 20:09:43 2025
 Verification finished: Wed Jun 25 20:14:23 2025
 MD5 checksum:    cbd7cb614fe75331f11649c9b2aad289 : verified
 SHA1 checksum:   ae6293f779910ba8d7eb12fb1aae717b80053733 : verified
```

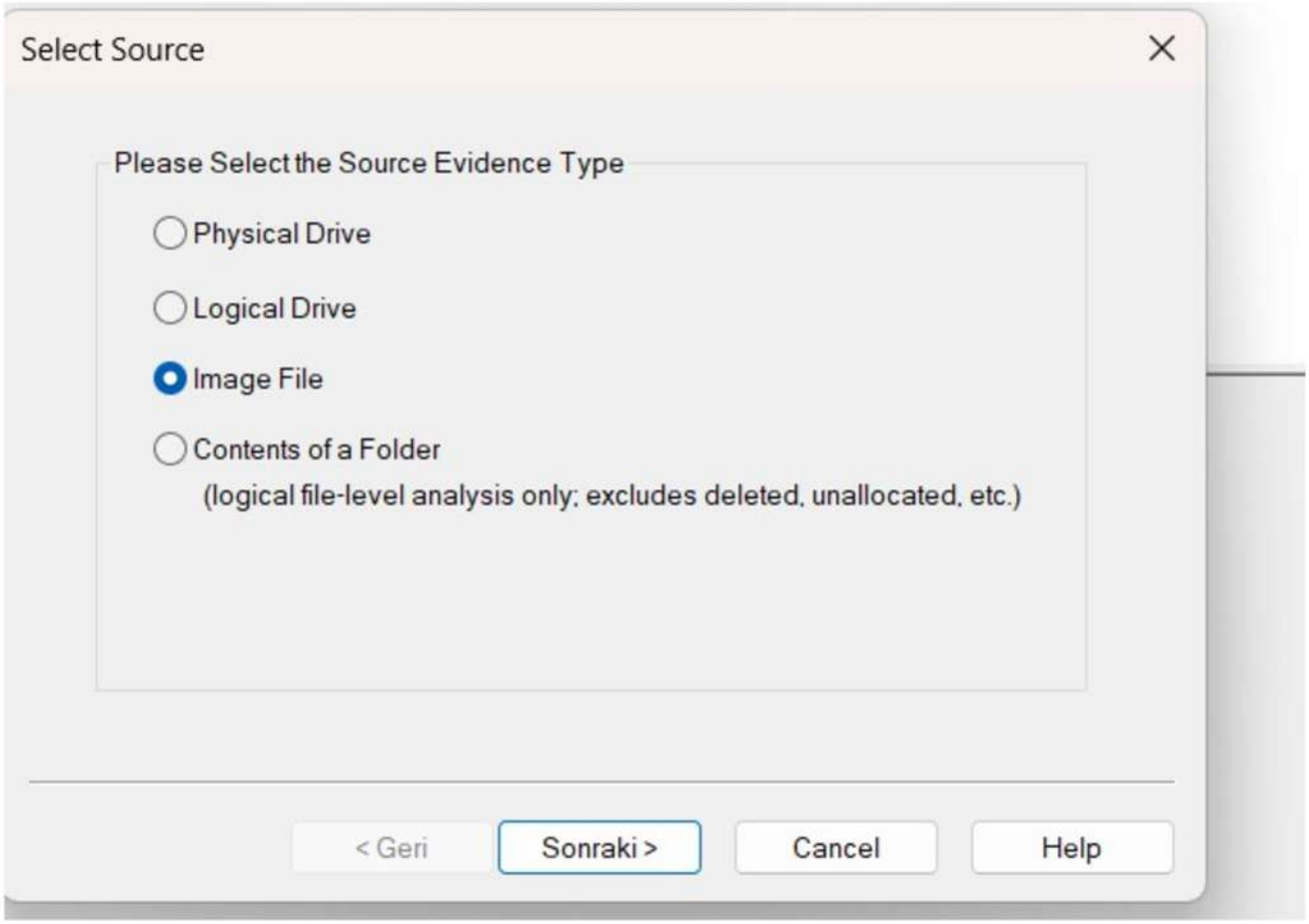


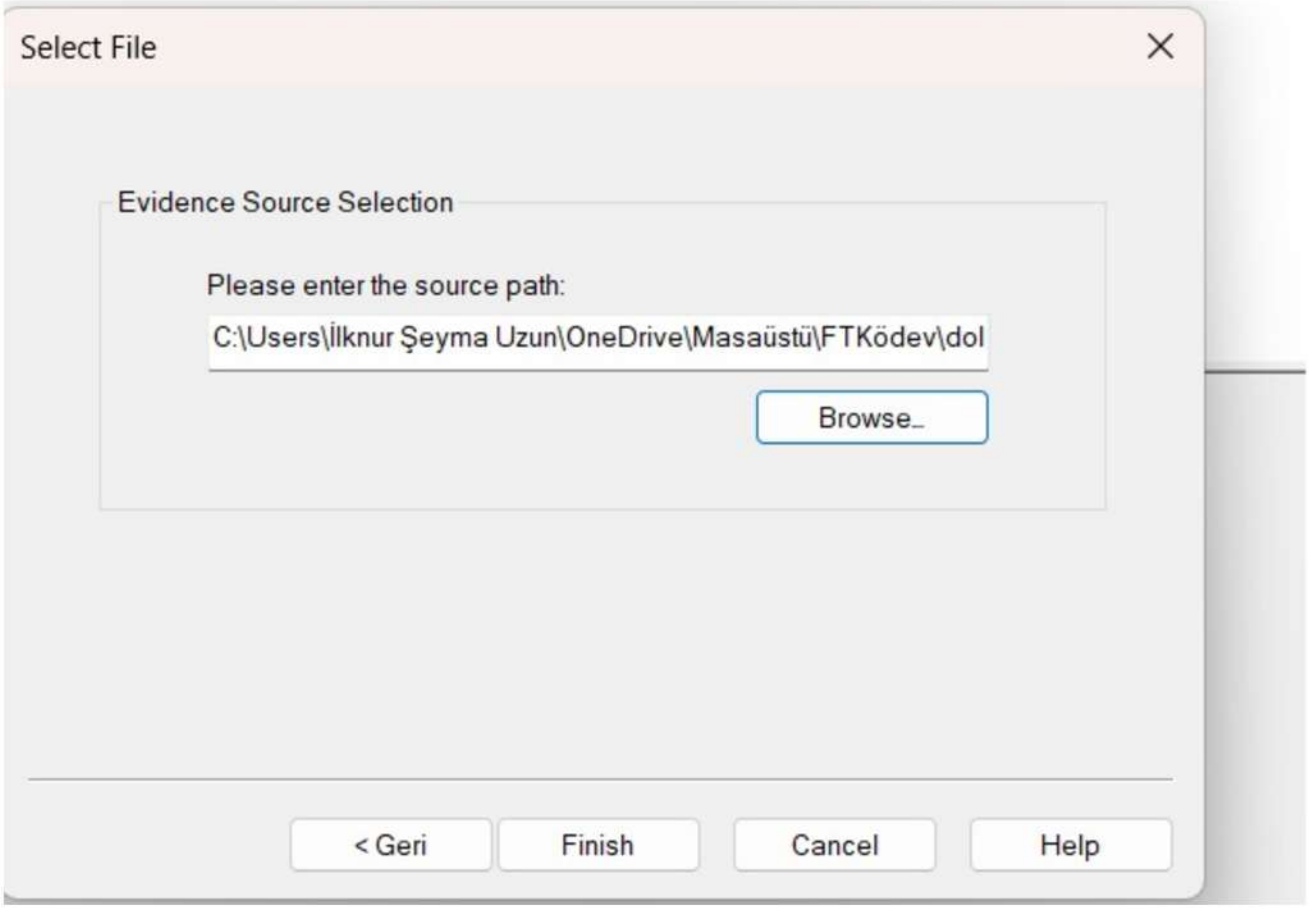
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Filename	Full Path	Size (bytes	Created	Modified	Accessed	Is Deleted						
2	[root]	Partition 1	16384				no						
3	VBR	Partition 1	512				no						
4	reserved s	Partition 1	1168896				no						
5	[unallocat	Partition 1	0				no						
6	FAT1	Partition 1	7803904				no						
7	FAT2	Partition 1	7803904				no						
8	System Vo	Partition 1	16384	2020-Aug-	2020-Aug-16 17:47:0(		no						
9	.Spotlight-	Partition 1	16384	2024-May	2024-May-03 10:24:3		no						
10	!UAWEI~1	Partition 1	16384	2021-Jun-2	2021-Jun-22 21:36:42		yes						
11	toplu	Partition 1	16384	2021-Jun-2	2021-Jun-21 11:02:46		no						
12	MELEK	Partition 1	16384	2021-Jun-2	2021-Jun-21 11:17:26		no						
13	melek albü	Partition 1	16384	2021-Jun-2	2021-Jun-24 22:19:18		no						
14	.fseventsd	Partition 1	16384	2024-May	2024-May-03 10:24:3		yes						
15	resimler	Partition 1	16384	2022-Aug-	2022-Aug-03 15:28:28		no						
16	Yeni klasö	Partition 1	0	2022-Aug-	2022-Aug-03 16:56:4:		yes						
17	mzk	Partition 1	16384	2022-Aug-	2022-Aug-03 16:56:4:		no						
18	gök foto	Partition 1	16384	2022-Sep-	2022-Sep-15 00:14:5(		no						
19	ÜNİVERSİT	Partition 1	65536	2022-Nov-	2022-Nov-25 17:25:3		no						
20	KİMYA NO	Partition 1	16384	2022-Nov-	2022-Nov-25 23:40:4		no						
21	DERS NOT	Partition 1	0	2024-Feb-	2024-Feb-07 23:42:42		yes						
22	untiy barb	Partition 1	16384	2024-Feb-	2024-Feb-05 23:41:0(		yes						
23	ilkVuforia	Partition 1	16384	2024-Feb-	2024-Feb-06 00:45:18		yes						
24	DERS NOT	Partition 1	16384	2024-Feb-	2024-Feb-07 23:42:42		yes						
25	ilkVuforia i	Partition 1	16384	2024-Feb-	2024-Feb-06 00:45:18		yes						
26	ilkVuforia	Partition 1	16384	2024-Feb-	2024-Feb-16 22:26:16		yes						
27	!roje	Partition 1	16384	2024-Feb-	2023-Dec-22 12:41:0(		yes						
28	proje ilknu	Partition 1	16384	2024-Feb-	2023-Dec-22 12:41:0(		yes						
29	!est	Partition 1	16384	2024-Feb-	2024-Jan-16 15:20:42		yes						

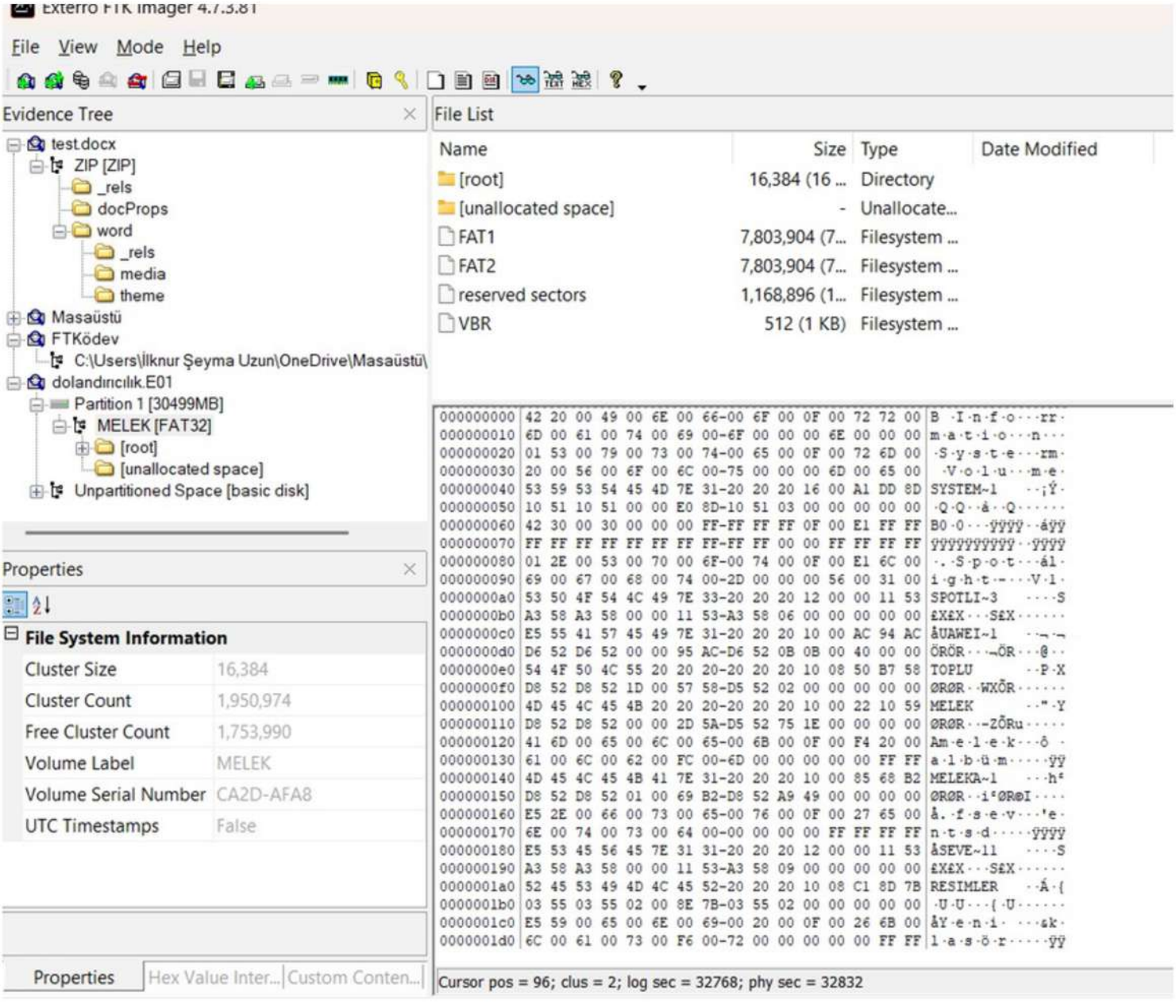
İmaj Dosyasını İnceleme

File > Add Evidence Item > Image File > Next > Browse > E01 dosyası > Finish
Sol panelde imaj eklendi

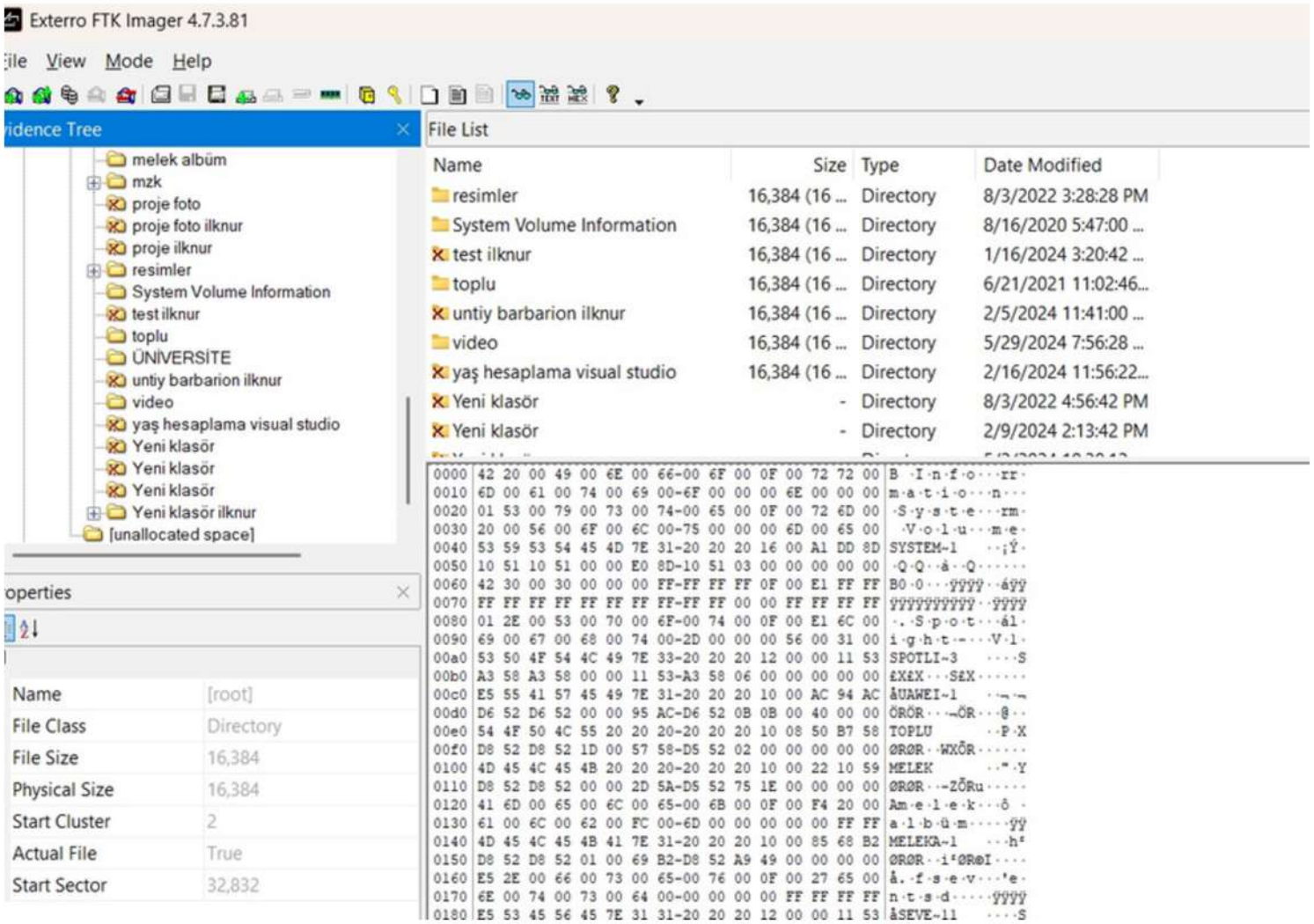





Select File
Evidence Source Selection
Please enter the source path:
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev\dol
Browse...
< Geri
Finish
Cancel
Help

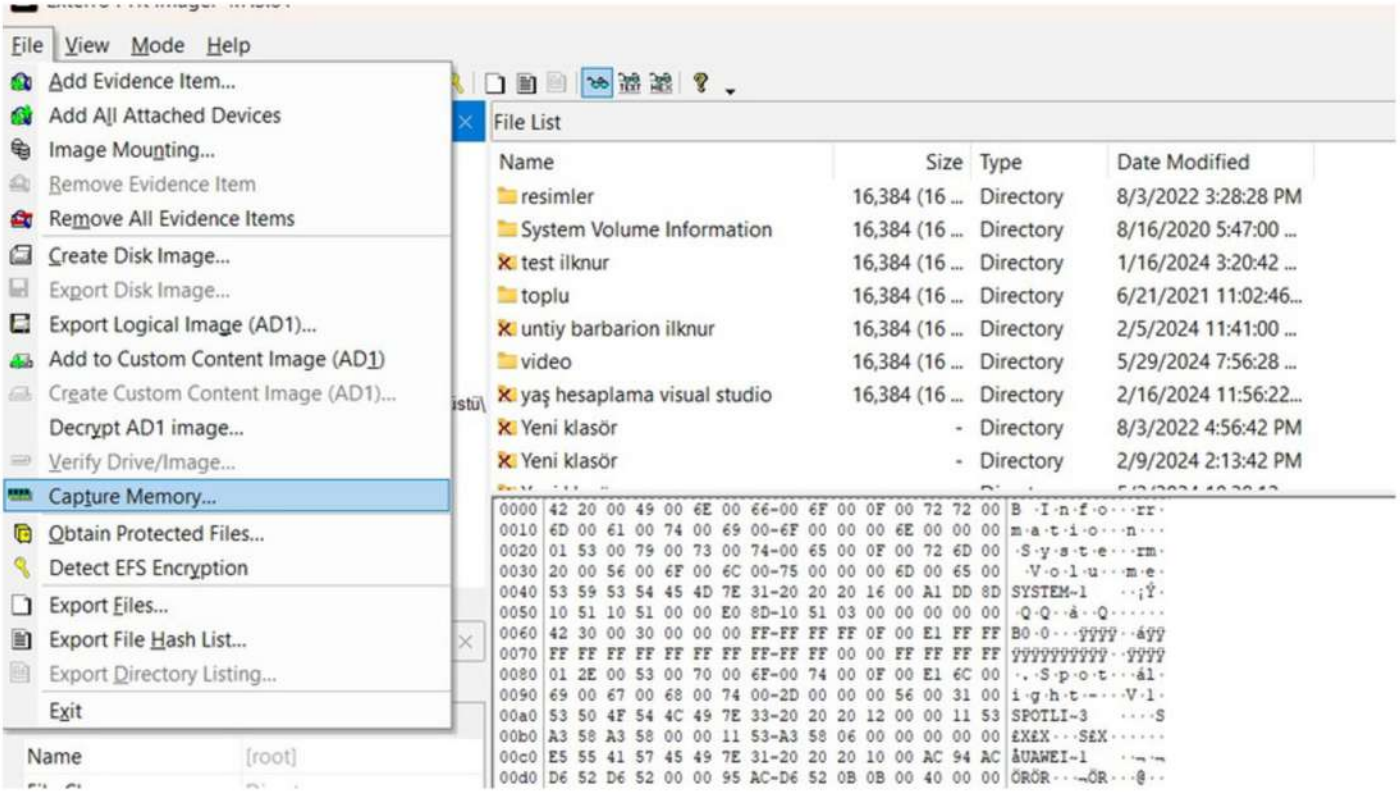

Exterro FTK Imager 4.7.3.81
File View Mode Help
Evidence Tree
test.docx
ZIP [ZIP]
_rels
docProps
word
_rels
media
theme
Masaüstü
FTKödev
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\
dolandırıcılık.E01
Partition 1 [30499MB]
MELEK [FAT32]
[root]
[unallocated space]
Unpartitioned Space [basic disk]
Properties
File System Information
Cluster Size 16,384
Cluster Count 1,950,974
Free Cluster Count 1,753,990
Volume Label MELEK
Volume Serial Number CA2D-AFA8
UTC Timestamps False
Properties
Hex Value Inter...
Custom Conten...
File List
Name Size Type Date Modified
[root] 16,384 (16 ... Directory
[unallocated space] - Unallocate...
FAT1 7,803,904 (7... Filesystem ...
FAT2 7,803,904 (7... Filesystem ...
reserved sectors 1,168,896 (1... Filesystem ...
VBR 512 (1 KB) Filesystem ...
Cursor pos = 96; clus = 2; log sec = 32768; phy sec = 32832

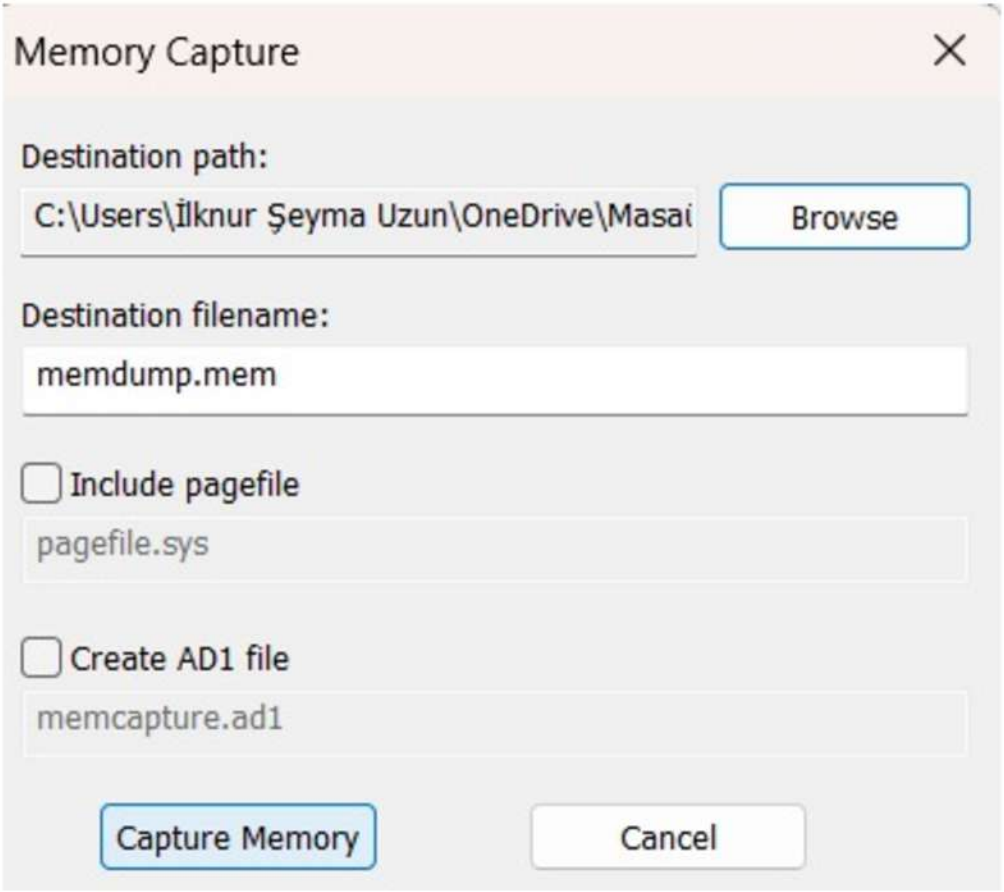
Burada silinen dosyaları görebiliriz ama kurtaramayız(silinen dosyalar için root a tıklayıp x işaretli gözükenler silinen dosalardır.) Kurtarmak için autopsy kullanılabilir.

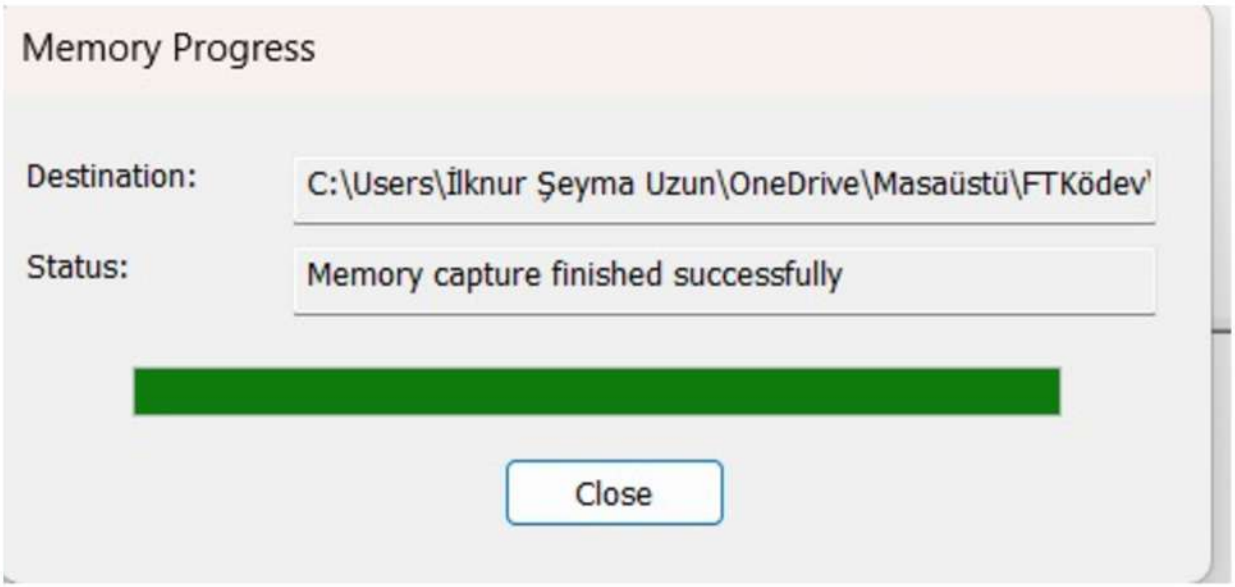


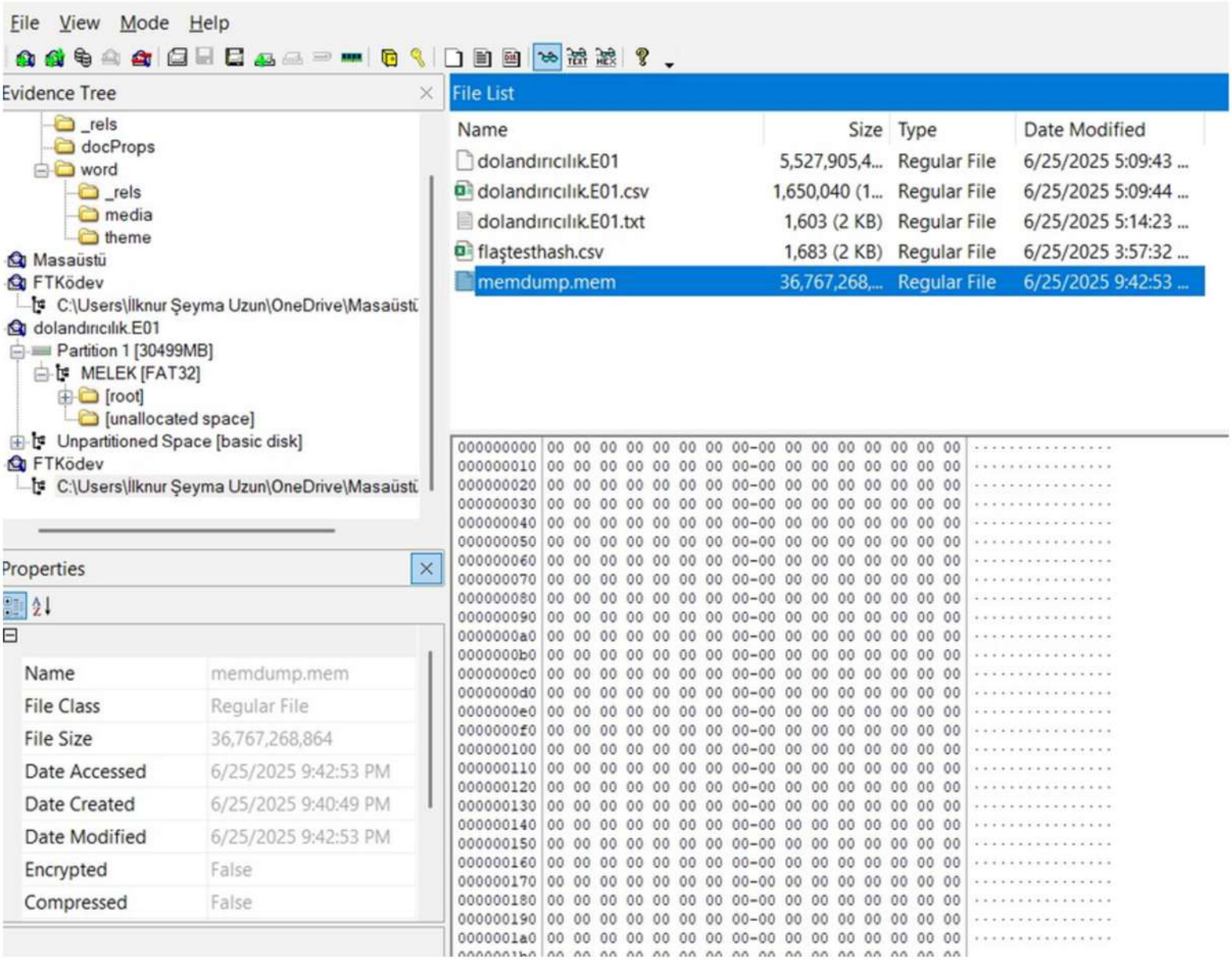
RAM Imajı Alma

RAM geçici bellektir bilgisayar çalışırken bilgileri burada tutar. Kapatıldığında RAM'deki tüm veriler silinir. Kapalıysa açmamak, Açıksa kapatmamak gerekir.






Memory Progress
Destination:
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü\FTKödev'
Status:
Memory capture finished successfully
Close


File View Mode Help
Evidence Tree
_rels
docProps
word
_rels
media
theme
Masaüstü
FTKödev
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü
dolandırıcılık.E01
Partition 1 [30499MB]
MELEK [FAT32]
[root]
[unallocated space]
Unpartitioned Space [basic disk]
FTKödev
C:\Users\İlknur Şeyma Uzun\OneDrive\Masaüstü
Properties
Name memdump.mem
File Class Regular File
File Size 36,767,268,864
Date Accessed 6/25/2025 9:42:53 PM
Date Created 6/25/2025 9:40:49 PM
Date Modified 6/25/2025 9:42:53 PM
Encrypted False
Compressed False
File List
Name Size Type Date Modified
dolandırıcılık.E01 5,527,905,4... Regular File 6/25/2025 5:09:43 ...
dolandırıcılık.E01.csv 1,650,040 (1... Regular File 6/25/2025 5:09:44 ...
dolandırıcılık.E01.txt 1,603 (2 KB) Regular File 6/25/2025 5:14:23 ...
flaştesthash.csv 1,683 (2 KB) Regular File 6/25/2025 3:57:32 ...
memdump.mem 36,767,268,... Regular File 6/25/2025 9:42:53 ...